Linux Final Notes

- Batch: Linux-B36
- Trainer- Abhipray Dhoble

Syllabus:

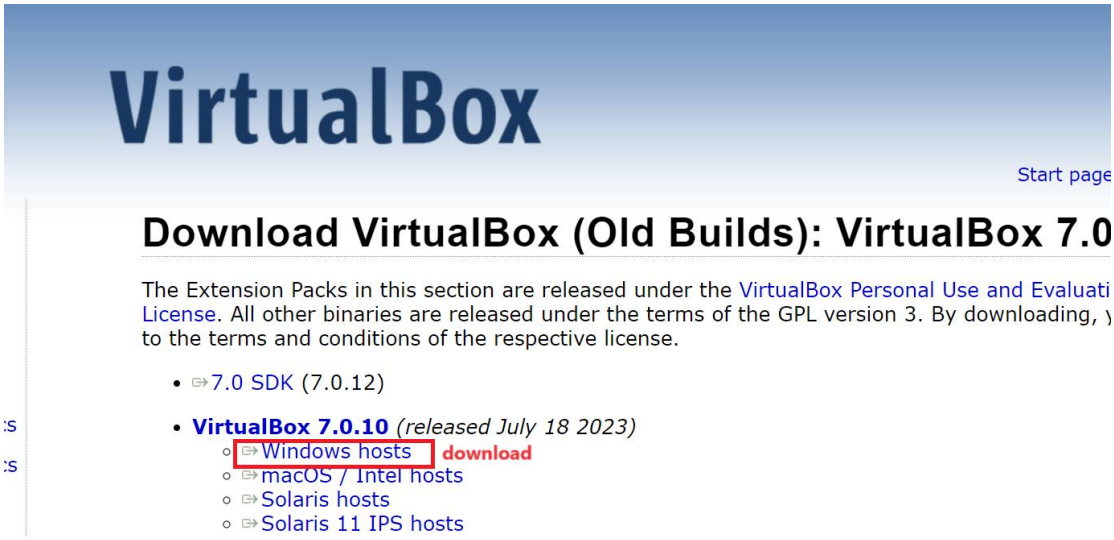
- Intro to Linux
- Basic commands
- Working with text Files and Directories
- Editors
- User and group management
- Files permissions
- Archiving and compression
- Filter and search utility
- Process management
- Package management
- Partition
- scheduling task
- package management
- Disk Partition
- Networking

- Web Server:
  ->Apache/Httpd
  ->Nginx
  ->Tomcat

- Services:
  ->SSH
  ->FTP
  ->NFS
  ->MariaDB
- Firewall

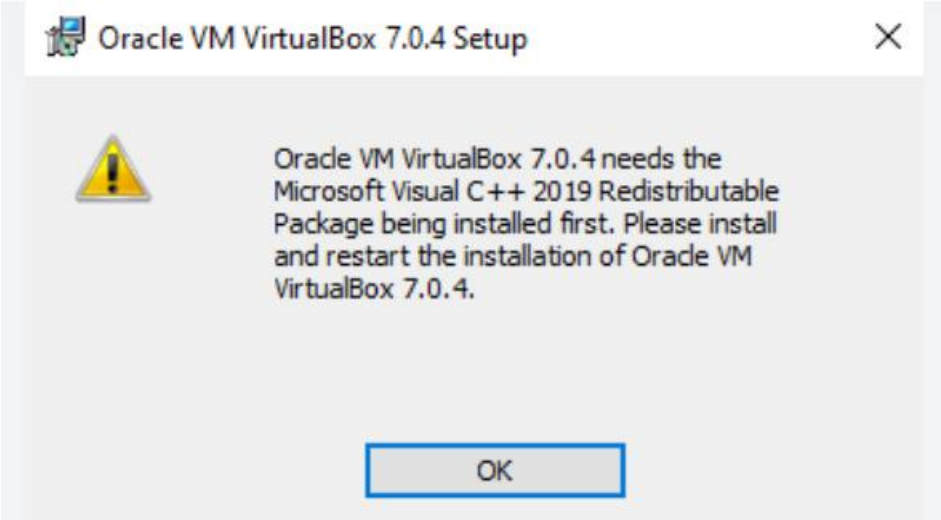
VM VIRTUAL BOX DOWNLOAD STEPS

Step1 -> Downoad Virtual box:
https://www.virtualbox.org/wiki/Download_Old_Builds_7_0



Step 2 -> install virtualbox

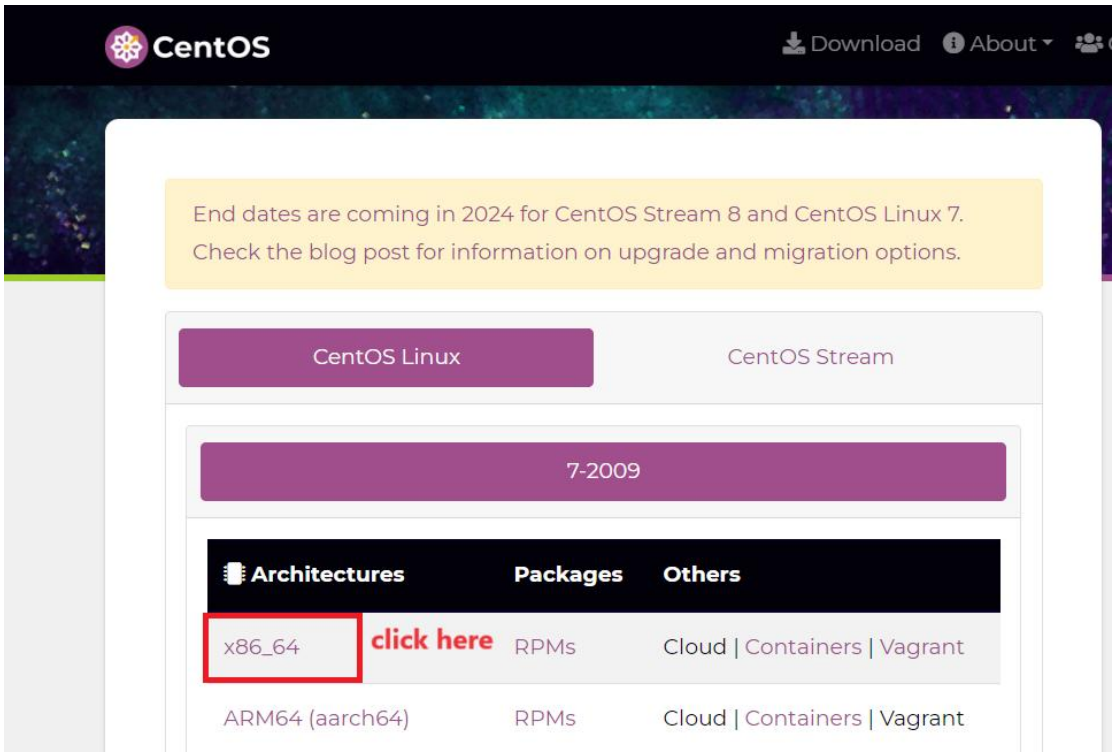
Note: if you get error like

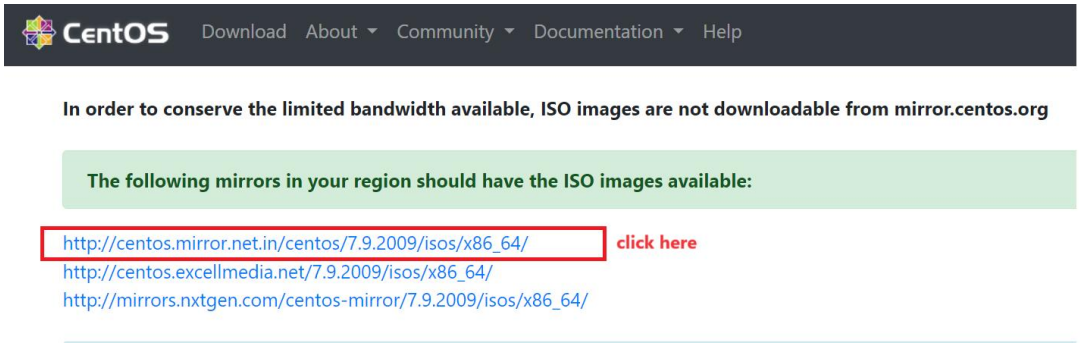


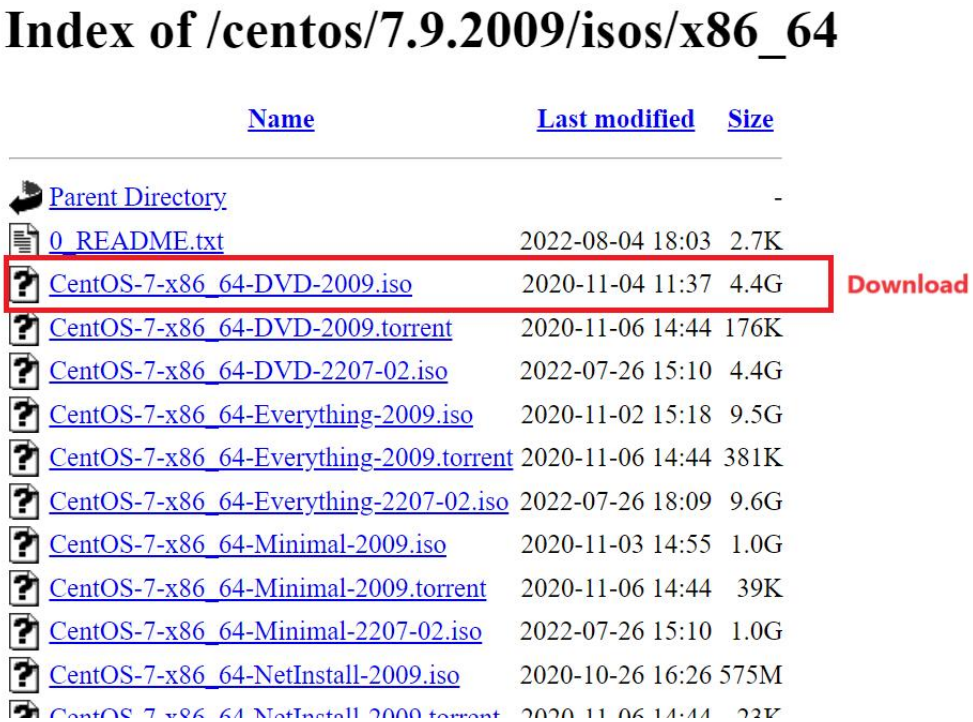
Visit:
https://learn.microsoft.com/en-us/cpp/windows/latest-supported-vc-redist?view=msvc-170

Architecture	Link	Notes
ARM64	https://aka.ms/vs/17/release/vc_redist.arm64.exe	Permalink for latest supported ARM64 version
X86	https://aka.ms/vs/17/release/vc_redist.x86.exe	Permalink for latest supported x86 version
X64	https://aka.ms/vs/17/release/vc_redist.x64.exe download and install	Permalink for latest supported x64 version. The X64

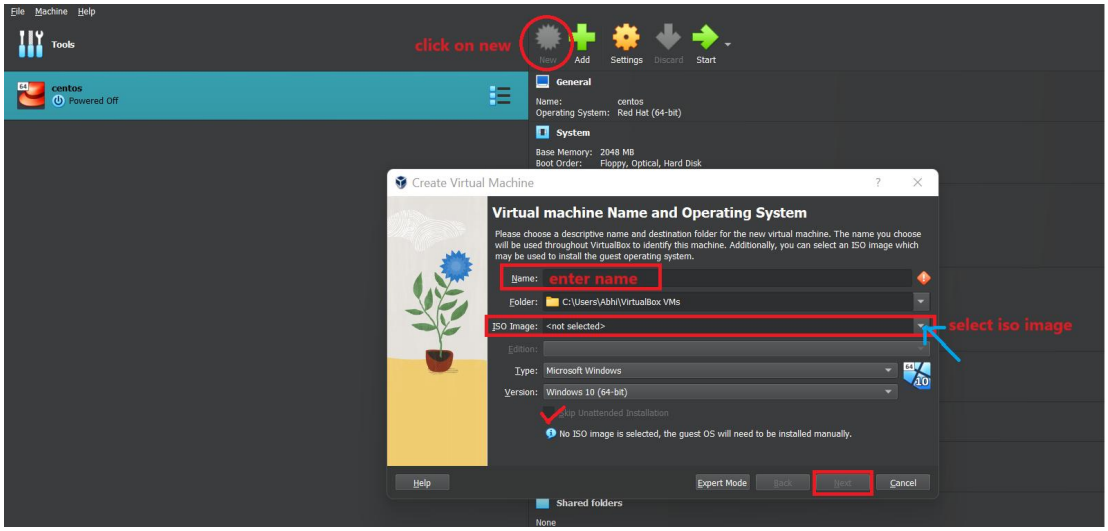
Step 3-> Dowmload Centos 7 ISO image:
https://www.centos.org/download/



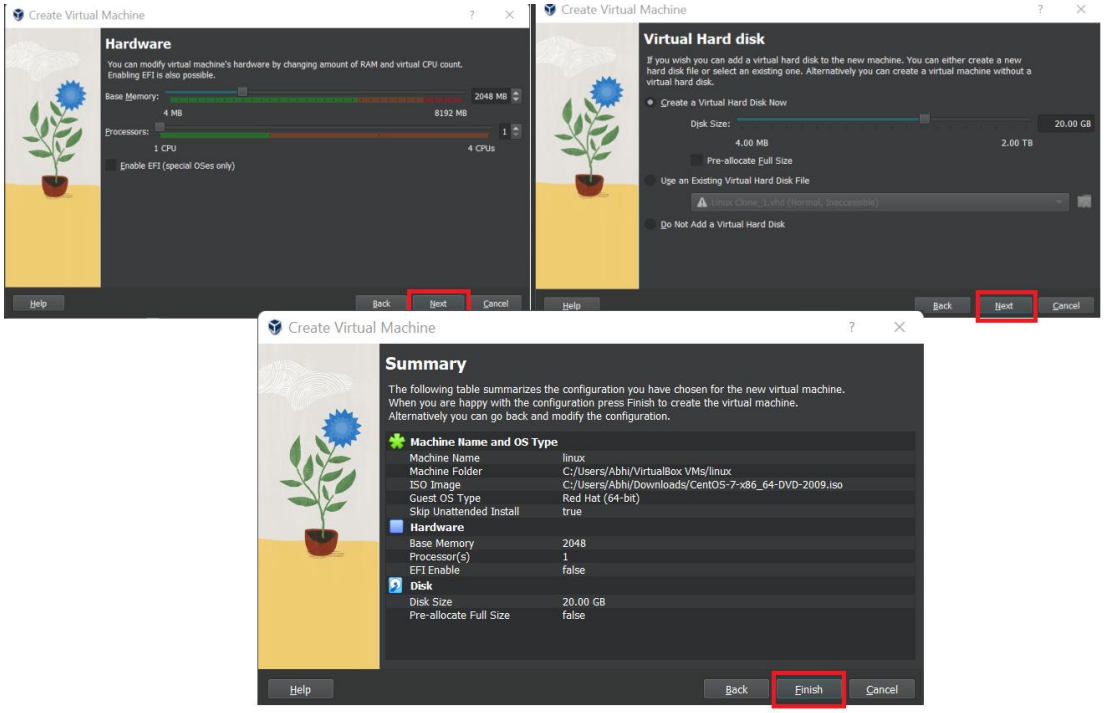


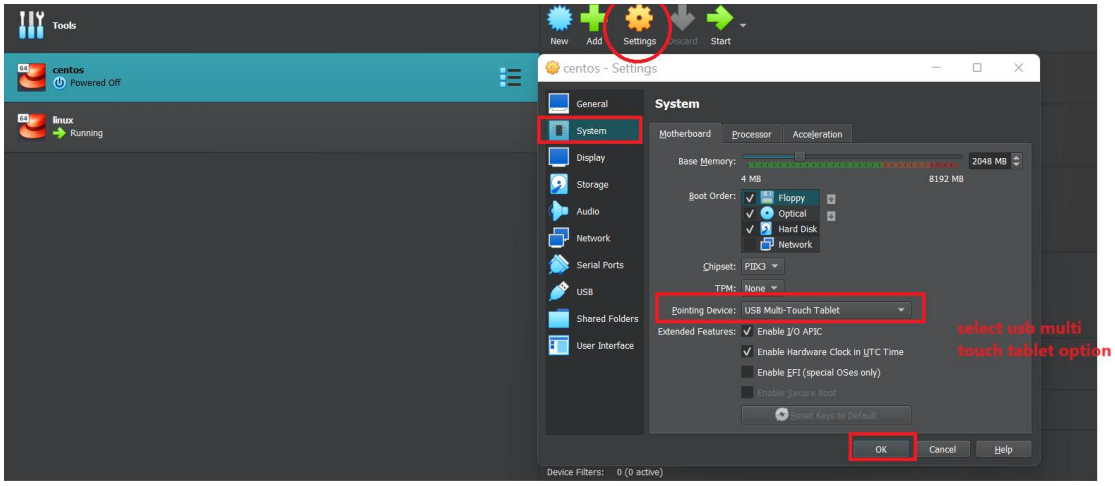


Step 4 -> Create Virtual Machine Inside VIrtual Box

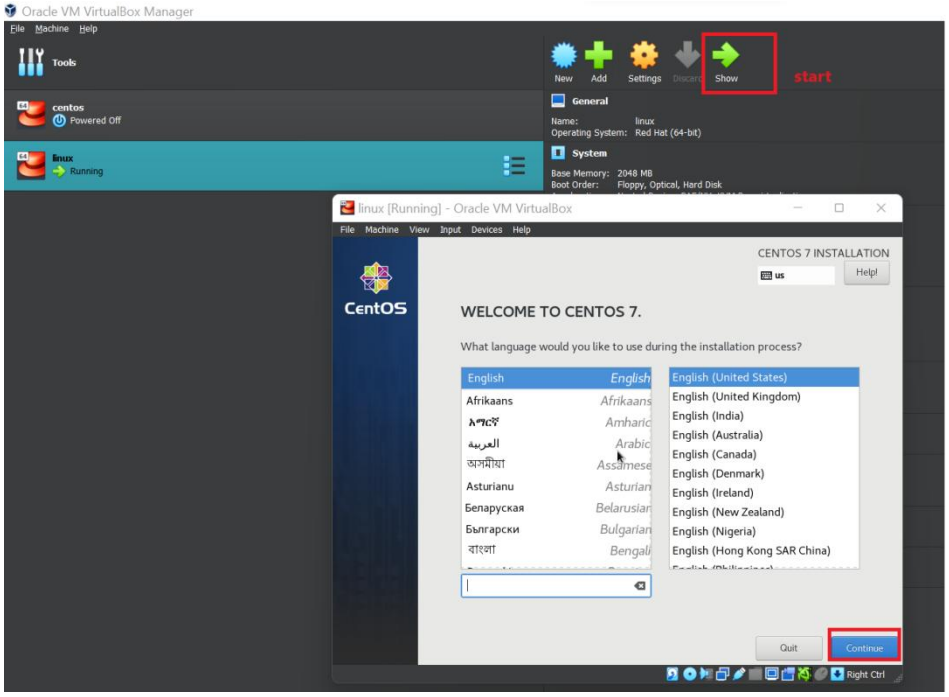


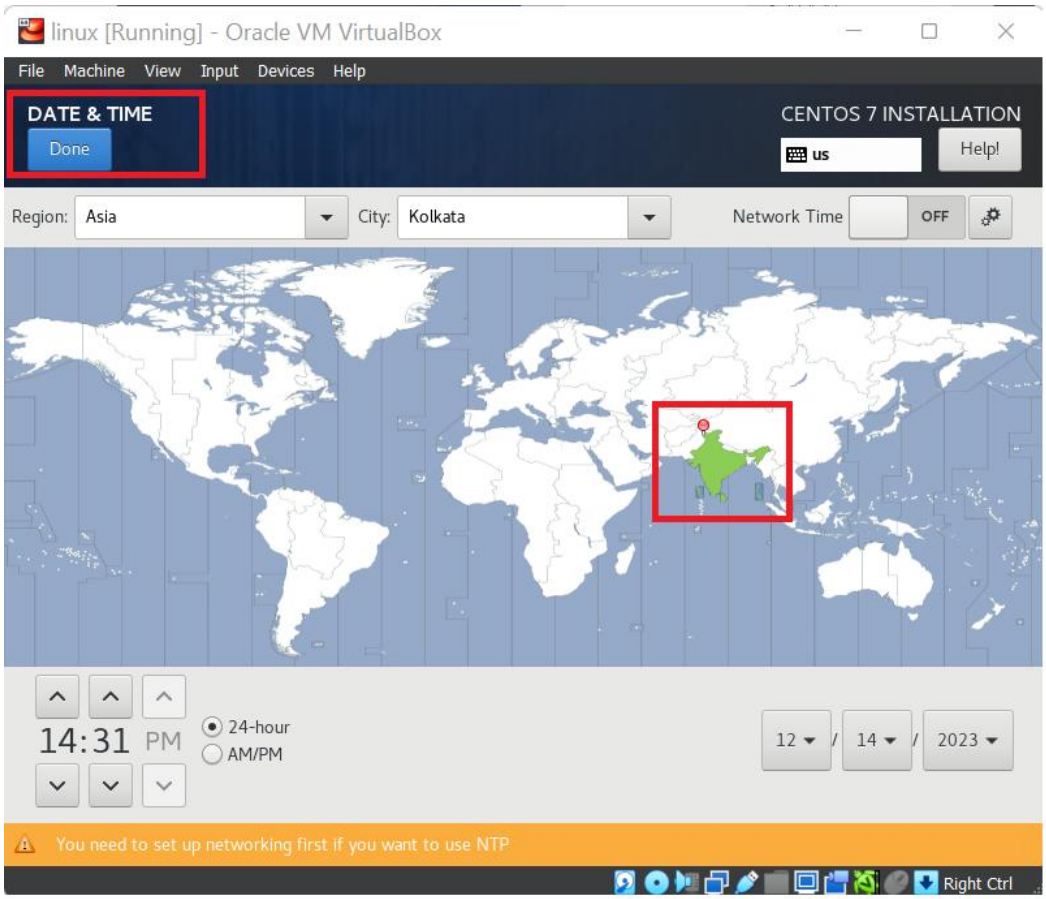
Step 5 : Do this setting to avoid mouse problem

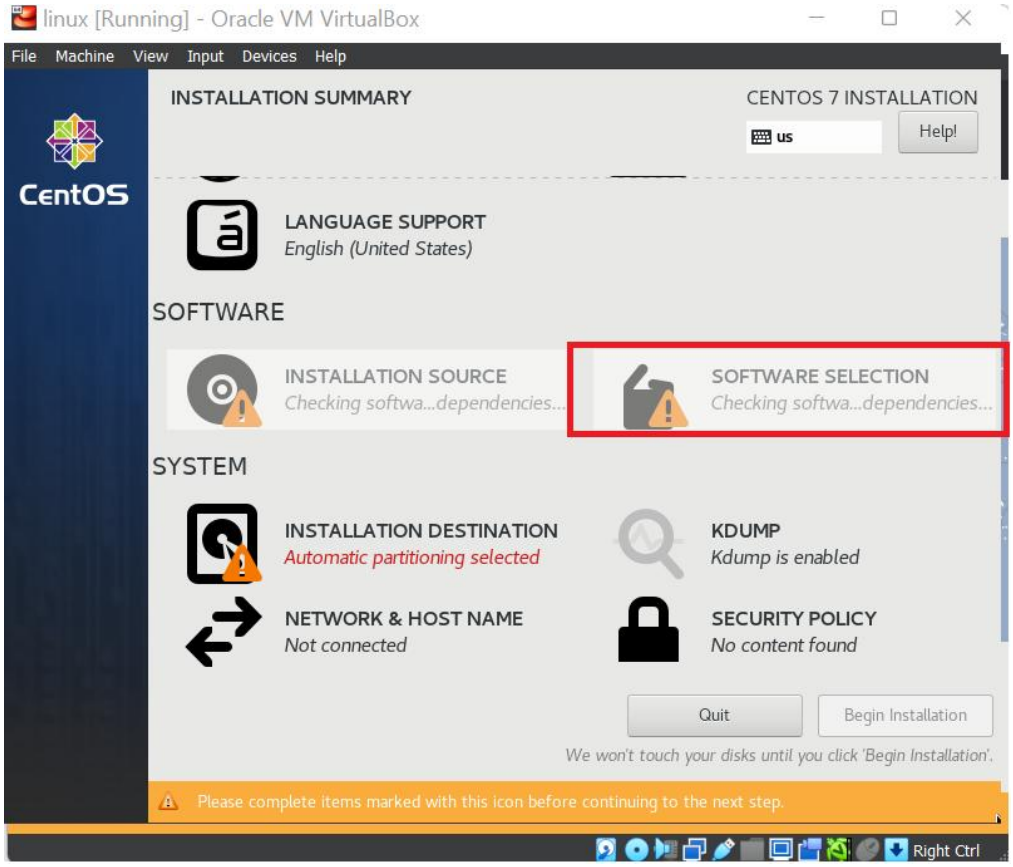


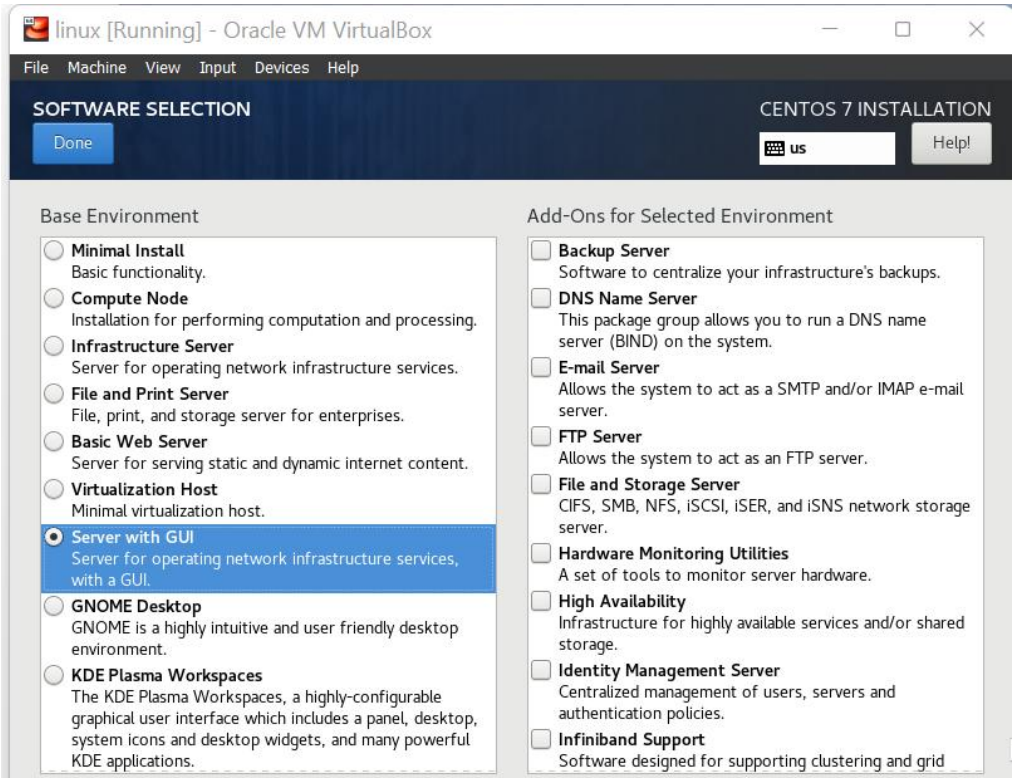


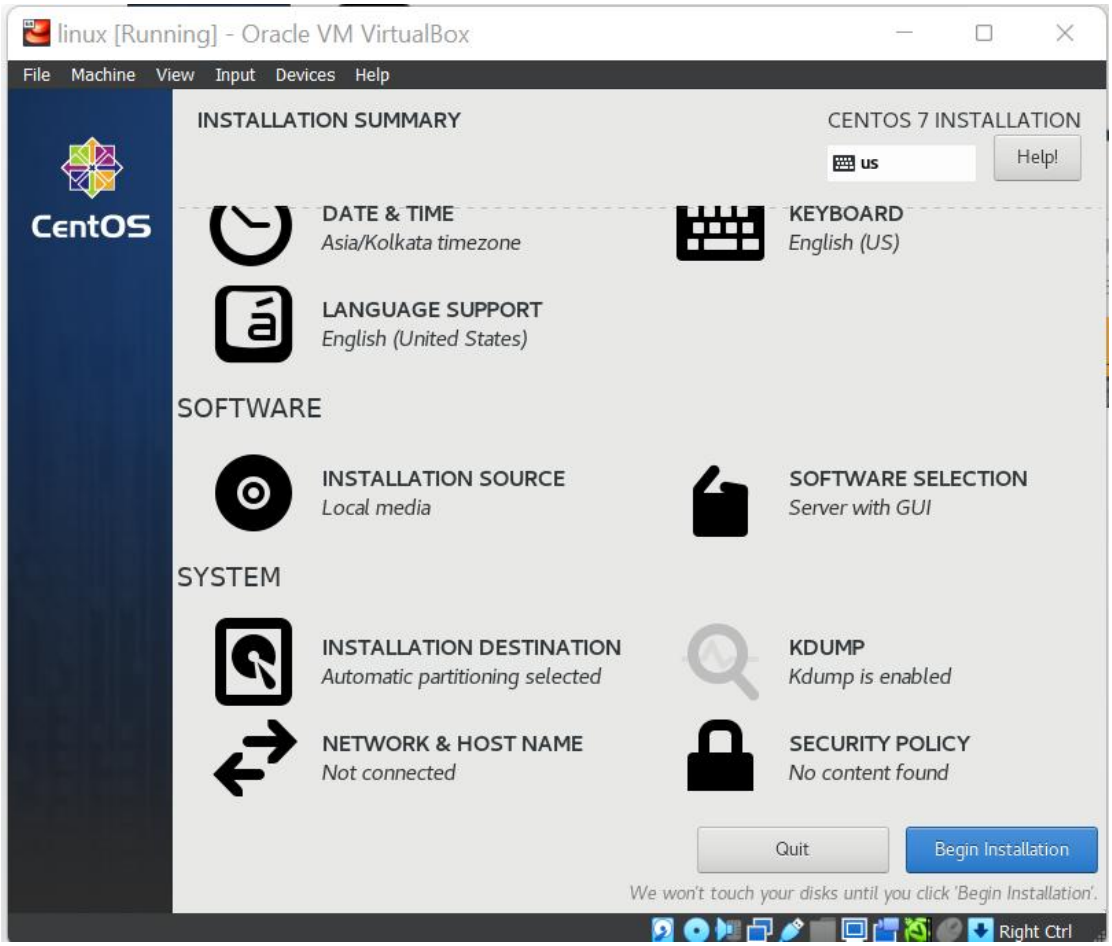
Step 6


Oracle VM VirtualBox Manager
File Machine Help
Tools
centos
Powered Off
linux
Running
New
Add
Settings
Show
start
General
Name: linux
Operating System: Red Hat (64-bit)
System
Base Memory: 2048 MB
Boot Order: Floppy, Optical, Hard Disk
linux [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
CENTOS 7 INSTALLATION
us
Help!
CentOS
WELCOME TO CENTOS 7.
What language would you like to use during the installation process?
English
Afrikaans
Amharic
Arabic
Assamese
Asturian
Belarusian
Bulgarian
Bengali
English (United States)
English (United Kingdom)
English (India)
English (Australia)
English (Canada)
English (Denmark)
English (Ireland)
English (New Zealand)
English (Nigeria)
English (Hong Kong SAR China)
Quit
Continue
Right Ctrl


linux [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
DATE & TIME
Done
CENTOS 7 INSTALLATION
us
Help!
Region: Asia
City: Kolkata
Network Time
OFF
14:31 PM
24-hour
AM/PM
12 / 14 / 2023
You need to set up networking first if you want to use NTP
Right Ctrl


linux [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
CentOS
INSTALLATION SUMMARY
CENTOS 7 INSTALLATION
us
Help!
LANGUAGE SUPPORT
English (United States)
SOFTWARE
INSTALLATION SOURCE
Checking softwa...dependencies...
SOFTWARE SELECTION
Checking softwa...dependencies...
SYSTEM
INSTALLATION DESTINATION
Automatic partitioning selected
KDUMP
Kdump is enabled
NETWORK & HOST NAME
Not connected
SECURITY POLICY
No content found
Quit
Begin Installation
We won't touch your disks until you click 'Begin Installation'.
Please complete items marked with this icon before continuing to the next step.
Right Ctrl


linux [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
SOFTWARE SELECTION
Done
CENTOS 7 INSTALLATION
us
Help!
Base Environment
Minimal Install
Basic functionality.
Compute Node
Installation for performing computation and processing.
Infrastructure Server
Server for operating network infrastructure services.
File and Print Server
File, print, and storage server for enterprises.
Basic Web Server
Server for serving static and dynamic internet content.
Virtualization Host
Minimal virtualization host.
Server with GUI
Server for operating network infrastructure services, with a GUI.
GNOME Desktop
GNOME is a highly intuitive and user friendly desktop environment.
KDE Plasma Workspaces
The KDE Plasma Workspaces, a highly-configurable graphical user interface which includes a panel, desktop, system icons and desktop widgets, and many powerful KDE applications.
Add-Ons for Selected Environment
Backup Server
Software to centralize your infrastructure's backups.
DNS Name Server
This package group allows you to run a DNS name server (BIND) on the system.
E-mail Server
Allows the system to act as a SMTP and/or IMAP e-mail server.
FTP Server
Allows the system to act as an FTP server.
File and Storage Server
CIFS, SMB, NFS, iSCSI, iSER, and iSNS network storage server.
Hardware Monitoring Utilities
A set of tools to monitor server hardware.
High Availability
Infrastructure for highly available services and/or shared storage.
Identity Management Server
Centralized management of users, servers and authentication policies.
Infiniband Support
Software designed for supporting clustering and grid



Step 7 -> Set Root Password

❖ Intro to Linux

what is Operating System

OS is a Mediator between application and hardware

- it provide environment to run services
- it provide platform to run application
- It manages hardware
- It is a set of Program (It converts High level language to low level language)

->High level language: human readable language (ex. Python, java, etc)

->Low level/Machine language: in the form of 0's and 1's(binary)

```
                    OS
user ------------------------> computer
          ( interface/mediator )
```

History of linux+

Linux is a kernel and it was written by Linus Torvalds in 1991

kernel + SOFTWARE = OS
(linux) (GNU) (LINUX)

GNU-->The GNU project is **a mass collaborative initiative for the development of free software**. founded by the Richard Stallman

What is Linux?

Linux an open source operating system (OS). An operating system is the software that directly manages a system's hardware and resources, like CPU, memory, and storage.

Open Source - Linux source code is freely available which means anyone can modify this code and create new version and it is community based development project. Multiple Teams works in collaboration to enhance the capability of Linux operating system

Architecture of Linux

- Linux architecture has four main components hardware,kernel,shell and user/application
- **Hardware :** it consists of motherboard ,CPU,HDD etc
- **Kernel :** kernel is the heart/core of the OS, kernel communicates with Hardware, It converts High level language to low level language
- **Shell** : provides interface to user to communicate with kernel
- **Application/user :** user interact with system via applications or commands

differences between windows and linux ?

Windows:

1.user friendly
2.licensed required {closed source-Microsoft }
3.requirement of hardware is more
4.single user
5.easily attacked by virus
6.low security
7.ram consumption high
8.FAT & NTFS file system
9.used in server and client machine

Linux:

1. not user friendly (command based ,skilled person required)
2. open source----anyone can develop
3. licence not required.
4. less requirement of hardware.
5. multiple users are allowed.(6 users can allowed on terminal)
6. highly secured .
7. ram consumption low
8. file system EXT & XFS
9. nearly 80% of Servers linux based

Que. Explain the FEATURES of Linux

1) Open source
2)secure
3)easy installation
4)light weight
5)multiuser
6)multiple distribution
7)portable
8)difficult to use
9)low hardware requirement

Various Distributions /Versions /Flavors of Linux

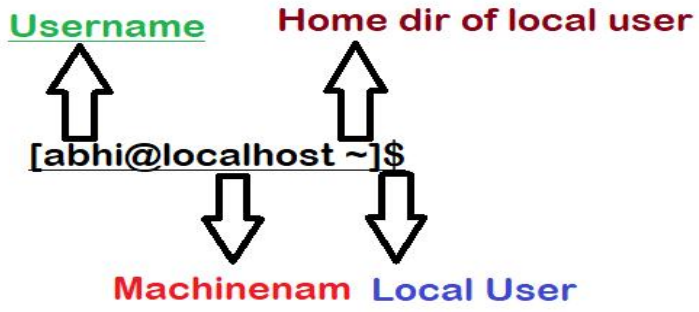
as Linux is open source OS therefore its source code is accessible to general public hence anyone can modify the source code and create a new version of OS

- RHEL(Red hat enterprise Linux)
- FEDORA
- DEBIAN
- UBUNTU
- KALI
- AMAZON LINUX
- CENTOS

Types of Users are there in linux

There are Three types of users in linux Local user, root and system user

- $ local user ->has limited privileges
- # root user or super user/administrator->has all the privileges
- system user -->automatically generated by system for a specific purpose or software.



❖ Basic commands

- whoami -->shows username
- who -->shows username with terminal, date and time

- ls --> list
- ls -l / ll --->longlist, check list along with permissions
- cd --> change directory
- man <command > -->shows manual page of command
- info <command> -->same as man command
- <command> --help --> shows options

- su root -->switch to root user
- sudo -i -->switch to root user
- sudo - root -->switch to root user
- su - username -->switch to another user
- exit -->logout from current user

- hostnamectl -->shows machine information like OS and kernel version etc.

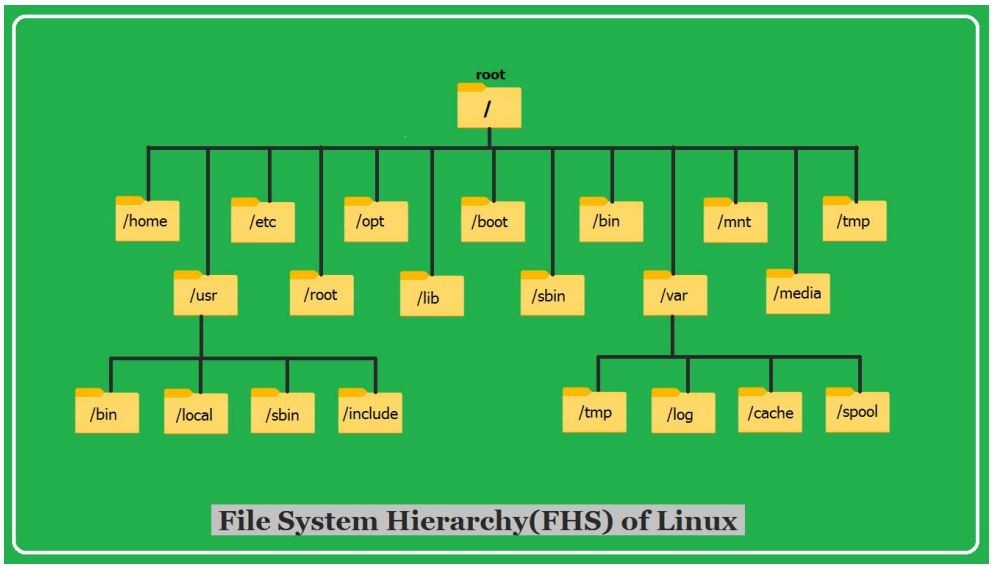
- hostname -->shows machine name
- hostname newhostname -->temporary change the hostname
  ->bash -->to save the changes made
- vi/etc/hostname -->permanently change the host-name
- cat /etc/os-release -->to check os version

- uname -->shows os name
- uname -a -->print system information like OS name, version, date etc

---

- dmidecode -->shows hardware info[user must have root permissions to run this command]
- free -h --> shows available memory [human readable]

Directory structure in linux



Directory structure in linux

Root /

- first directory in structure
- Every files and directory starts from the root directory.
- Only the root user of the system has the right to write under the root (/) directory.

*One another directory with name root is under the root directory (/root) is the home directory of the root user.

1)/bin – local user's Binary files[not human readable]
2)/boot –It stores all information about boot loader.Kernel initd, grub files are located into /boot directory.
3)/dev – [Device Files] It includes terminal devices, USB, or any device attached to the system. ex./dev/tty1, /dev/usb

4)/etc – System and program configuration files

5)/home – It contains the user's home directories,
personal files, user's personal settings, etc.
ex. /home/abhi

6)/lib &/lib64 – [Shared Libraries] It stores libraries essential files for
the binaries in /bin and /sbin.

7)/media – It is temporary mount point for removable device such as
/media/cdrom, /media/floppy

8)/mnt – It is for a temporarily mounting filesystem, where system
admin can mount a file system

9)/opt –used for optiona application software packages

10)/root -it is the home directory of the root user.

11)/run -it stores data that can give you an idea of how system
resources are being utilized since startup

12)/sbin – Root user's binary files,
It stores system related binaries executable. ex.ifconfig

13)/srv –service
it contains server specific services related data

14)/sys - it contains information about various system components and
drivers

15)/tmp – It stores files which are created by system or user for a
temporary purpose.

16)/usr – [bin, sbin] It contains libraries, binaries, documentation, and
source code

17)/var - varible
Stores files whose content is expected to change continuously
ex. log files, spool files and cache files.

18)/proc –stores information about system process information
ex. w, top, uptime

❖ Working with text Files and Directories

Que. How to create directory ?
-->mkdir <foldername>
ex. mkdir mycomputer

Que. How to remove empty directory ?
-->rmdir <foldername>

Que. How to remove directory with content?
-->rm -rvf <filename/foldername>

r ->recursive[delete folder/file within file/folder]
v ->view/verbose[shows the action taken on screen]
f ->forcefully remove

Que. How to make multiple folder
-->mkdir <folder1> <folder2>

Que. How to make folder inside another folder
-->mkdir <foldername>/{fold1,fold2}

Que. How to make sub directories within main directory
-->mkdir -p <foldername>/dir1/dir2/dir3
ex. mkdir -p Asia/India/Maharashtra/Nagpur

Que. How to create 20 directories at once
-->mkdir <mydir>{1..20} --->it will create 20 dir with the name of folder1 to folder20

Que.how to make empty file ?
-->touch <filename>
ex. touch abc.txt

Que. How to create file using cat command ?
-->cat > filename
->Hello world!
*enter
*press cntl + d to save the file

Que.How to append [edit] the content in the file using cat ?
-->cat >>filename

Que. How to view content in the file ?
-->cat filename

Que.How to reverse the content of the file [bottom to top]
-->tac filename

Que.How do I simultaneously view the content of two diff files
-->cat file1 file2

Que. How to merge the content of two files into another one? Or copy
-->cat file 1 file2 >>newfile

❖ Editors

Que. What is vi/vim and explain modes of vi?
-->Vi/Vim is a CLI (command line interface) based text editor in Linux
there are total 4 modes available in vi/vim

1.Command Mode:
>**this is default mode**. Press **esc** to exit from any mode and
enter into command mode.

dd Delete current line
<n>dd Delete n no. of lines from current line
yy Copy current line
<n>yy Copy n no. of lines from current line
cc Cut current line and enter in insert mode
<n>cc Cut n no. of lines and enter in insert mode

yw Copy current word
<n>yw Copy n no. of words from current word
dw Delete current word
<n>dw Delete n no. of words from current word
cw Cut current word and enter in insert mode
<n>cw Cut n no. of words from current word and enter in insert mode
p Paste
u Undo
Ctrl+r Redo
H Move cursor to the top of screen
M Move cursor to the middle of screen
L Move cursor to the bottom of screen
G Move cursor at the end of file
gg Move cursor at the beginning
<n>gg Move cursor at nth line

2.Insert Mode:
Following are commands to enter in insert mode

i insert text at current cursor position
I insert text at start of the current line
a insert text just right of the current character

A insert text at end of the current line
o insert new line below the current line
O insert new line above the current line
r it replaces single character
R replace multiple characters

3.Ex-mode

:q quit without saving
:q! quit without saving forcefully
:w save and stay in file
:wq or **:x** save and quit
:wq! save and quit forcefully
:set nu set line numbers
:<n> Jump to nth line
:set nonu remove line numbers
:/<word> Highlight word
:nohl Remove highlight
:%s/<old>/<new>/g Find and replace old word with new word
:!<command> Execute any command on terminal without leaving editor

4.Visual Mode: this mode is used for selection.

v Select character by character
V Select line by line
ctrl+v Select block
y,d,c For copy, delete, and cut selected area

Que. How I can execute command inside vi editor
-->:! ls

; [semicolon] execute two or more commands at once

| [pipe] it matches 1st command output to second command and executes it

Que. What are the Re-directors used in linux?

> write content in the file [override]
>> add new lines along with older ones [append]
2> save error in file
2>> save error in same file [append]
&> save output + error
&>> append

❖ User and group management

Users:

1. *Local user $* [normal user]

- Need to access files and folders stored on the computer.
- Has a limited access to computer resources

2. *Root user #* [super user/administrator]
- **The root account has all the *privileges*.** This means it can read and write any files on the system, perform operations as any user, change system configuration, install and remove software, and upgrade the operating system

3. *System user*
- created by system itself when we install any service to perform necessary activities where no human intervention needed
- It has no-login shell

Que.How to add new user?
-->useradd username
-->adduser username

Que. How to set password to user?
-->passwd username
>this command will help you to set password for user
Note: if you want to set password for current or root user use #passwd

Que. How to delete user?
-->userdel username [it will only delete user]
-->userdel -r username [it will remove user with home dir]

Que. What happens when we add new user to system?

1. **Home/username** >home directory of user created

2. copy files from **/etc/skel/** to **/home/username/**
   .bashrc ------> helps to login into the system
   .bash_logout ------> helps to logout from the system
   .bash_profile ------> creates Home directory of user

Note: to view above files use ls -a (shows hidden files)
#ls -a /home/username

3. **/etc/passwd** ------->it stores profile related info of user
   #tail -1 /etc/passwd

4. **/etc/shadow** ------> it consist of user password

5. */etc/group* ------->*group info such as group name,gid & list of members*
6. */etc/gshadow* ------->*it consists of group password*

Note: when you add a new user in system,the primary group will be created with the same name as user

7. **/var/spool/mail/username** ---->all the user's mails will be stored here

Que. What are the Fields of /etc/passwd ?

```
[root@centos /]# tail -1 /etc/passwd
diana:x:8183:8185::/home/diana:/bin/bash
```

[username:X:8183:8183:comment:/home/username:/bin/bash]
1.username
2.encrypted password
3.uid
4.gid
5.comment
6.path of home directory
7.login shell

Que. How to add user with customization:

Syntax-
 # useradd <options> <parameters> <username>

-->useradd -u 2023 user1
>to assign customized **uid** while user creation

-->useradd -m -d /Dev/user3 user3
>assign diff home directory

-->useradd -s /sbin/nologin user4
>assign no login shell to user[user wont be able to login]

-->useradd -c “development ” user5
>add comment

-->useradd -M Barry
>home directory of user will not be created

Que.How to modify existing user with customization:
Syntax-
 # usermod <option> <parameters> <username>

-->usermod -u 2023 user1
>to assign customized **uid** while user creation

-->usermod -m -d /Dev/user3 user1
>assign diff home directory

-->usermod -s /sbin/nologin user1
>assign no login shell to user[user wont be able to login]

-->usermod -c “development team” user1
>add comment
-->usermod -M user1
>home directory of user will not be created
-->usermod -L user1
>lock user password

```
[root@centos /]# usermod -L diana
[root@centos /]# tail /etc/shadow | grep diana
diana:!$6$zg3Wentk$9/ADy3K7pB7dhPCdM8SC/B0.HO56242LlGvs2gAyFl94LDO4xN3oEC5UZNZqjB0dDdPZ
p/R8NqTKaivUu3/mZ1:19570:2:99999:7:5:19589:
```

#usermod -U user1
>unlock user password

```
[root@centos /]# usermod -U diana
[root@centos /]# tail /etc/shadow | grep diana
diana:$6$zg3Wentk$9/ADy3K7pB7dhPCdM8SC/B0.HO56242LlGvs2gAyFl94LDO4xN3oEC5UZNZqjB0dDdPZp
/R8NqTKaivUu3/mZ1:19570:2:99999:7:5:19589:
```

Que. Explain Fields of /etc/shadow file?

```
[root@centos /]# tail -1 /etc/shadow
diana:$6$zg3Wentk$9/ADy3K7pB7dhPCdM8SC/B0.HO56242LlGvs2gAyFl94LDO4xN3oEC5UZNZqjB0dDdPZp
/R8NqTKaivUu3/mZ1:19573:2:5:7:5::
```

1 username
2 encrypted password
3 last password change
4 minimum days between password change
5 maximum days between password change
6 warning for password change
7 inactive days
8 expiry date
9 future use

Que. Explain password policy ?

-->A password policy defines the password strength rules that are used to determine whether a new password is valid

->chage -l
-l list or view
-m min days > chage -m 3 username
-M max days > chage -M 15 username
-I inactive days > chage -I 7 username
-E expiry date > chage -E '2023-9-1' username
-d force to change password
-W warning

Change password policy:

Syntax-
 # chage <option> <parameters> <username>

#chage -l username --to list pass policy
#chage -m 2 username --min days between pass change
#chage -M 20 username --max days between pass change

#chage -I 5 username --set inactive days for user
#chage -W 7 username --set warning days to change pass
#chage -E '2023-8-15' username --set expiry date for user

Groups:

> Group is the collection of users

Que.what are the Fields of /etc/group ?

```
[root@centos /]# tail /etc/group
```

```
jsa:x:2012:bruce,diana,clark
```

- Group name
- Redirected group password
- Group id
- List of members in the group

Que. Exxplain the Fields of /etc/gshadow

```
[root@centos /]# tail /etc/gshadow
```

```
jsa:$6$HEfBF/41ce4inhIH$tKhzFQc9ea3cbVp0vejfNoQriS0/AU1DSPutEdMyw0vtowy
VEGDB2v.9ktPqmP4ykYfL0iXsqQJ8qPkblRV0l.:diana:bruce,diana,clark
```

- Group name
- Encrypted group password
- Group admin
- List of members in the group

Que.How to create group?
-->groupadd groupname

Que.How to delete group?
-->groupdel groupname
-->groupdel -f groupname ---for forcefully delete group

Que.How to Add user to group?
1>#usermod -aG groupname username
 >**to add already existing user to group**

2>#useradd -G groupname username
 >to create new user and add it to the group

3>using gpasswd:
Syntax-
 # gpasswd <option> <username> <groupname>

#*gpasswd -a username groupname*
>**to add already existing user to group**
#add multiple users to group
#gpasswd -M mangesh,shubham dev

Que.How to delete user from group?
#gpasswd -d username groupname

Que. How do you Assign Admin to group?
#gpasswd -A username groupname

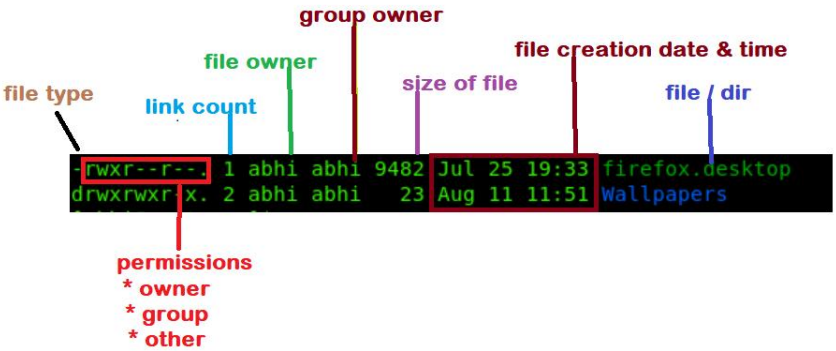
Que.How do you Remove Admin from Group?
#gpasswd -A ‘ ’ groupname

Que. What is the user id of a root user ?
-->By default **root user id is 'o'**

❖ permissions

- Permissions determine who can access files
- And specify who can read, write, modify files/directories on a system

Note : Use ls -l command to check File/Dir permissions
>ll
>ll -d dirname ---to check particular Dir permission
>ll filename ---to check particular file permission
>ls -ltr



1. File type
2. Owner Permissions.
3. Group Permissions.
4. Other User Permission.
5. Link Count.
6. Owner of file/directory.
7. Group Owner of File or Directory.
8. File Size.
9. Creation Date and Time.
10. File/Directory Name

File types in Linux:

- Normal file >text files, binary files etc..
d Directory
l Link file

Link Count: It shows count of links of file/directory that has been created.
>default link count for Dir =2
>default link count for File =1

Diff between Hard link and soft link

Hard Link	Soft Link
1.Inode number of hard link is same as that of inode number of original file.	1.Inode number of softlink is different than inode number of original file.
2.size of hardlink and original file will be same.	2., size of softlink will be different
3.Creating/removing hardlink will increase/decrease link count by 1	3.Creating / removing softlink will not affect link count.
4.hardlink of directory is not possible.	4.softlink of directory is possible.
5.ln <original_file> <hardlink_name>	5. ln -s <original_file> <hardlink_name>

Inode number: inode number is generated to keep track of all the files on a Linux system

User and group ownership:
owner --->current user, who created the file
Group --->primary group of file owner
other --->other users

read r 4 --->open or view/list
write w 2 --->edit
execute x 1 --->run

change owner and group owner

Lets take an example, "demo" dir owner and group is *root*

```
drwxr-xr-x. 2 root root 6 Aug 12 10:10 demo
```

Now I want to make user *abhi* the owner of the dir "demo"for that use the following command

#chown abhi demo --->change owner

```
drwxr-xr-x. 2 abhi root 6 Aug 12 10:10 demo
```

Same with the group

#chgrp grp-name demofile.txt --->change group owner
#chown user:Grp demofile.txt ---->change owner,grp at once

Giving permissions to user/group/others individually:

chmod u+w demofile.txt --->add user permission to write
chmod u-w demofile.txt --->remove
chmod g+rwx demofile.txt --->add rwx perm. to grp owner

chmod g-r,o=rw demofile.txt --->
chmod go=r dir1 --->assign grp and owner read permission

Giving permissions to u-g-o at once :

chmod ugo+rw demofile.txt
or
chmod a+rwx demo.txt --->a-all at once

octal vs binary

Octal	Binary	Permission
1	001	--x
2	010	-w-
3	011	-wx
4	100	r--
5	101	r-x
6	110	rw-
7	111	rwx

Default Permission for root and local user

User	Directory	File
Root	755	644
localUser	775	664

Giving permissions to user/group/others using octal number:

```
#chmod 421 /abhi/
#ll -d /abhi/
dr---w---x  2 root root 11 aug 23  07:10  /abhi/

#chmod 642  demofile.txt
-rw-r--w-   1 root root  11 aug 23  07:11  demofile.txt
```

Umask

-->umask is used to set default permissions for newly created files and directories

Calculate umask:

full permission - default permission =umask

>For root user default umask is 022

777 - 755 =022 root -->dir
666 - 644 =022 root -->file

>For local user default umask is 002

777 - 775 =002 local user -->dir

666 - 664 =002 local user -->file

#umask [to check current umask]
#umask 000 [to set umask value temp]
#vi /etc/bashrc or /etc/profile [permanently set umask for all users]
#source /etc/bashrc [to apply changes]

Calculate umask:
0 : read, write and execute
1 : read and write
2 : read and execute
3 : read only
4 : write and execute
5 : write only
6 : execute only
7 : no permissions

Ex. Suppose you want to set umask 013 which means u-g-o will have following permissions
User --->read, write and execute
Group --->read and write
Others --->read only
(max perm) 777 - 013 (umask) =764 [rwx rw- r--]

---

special permission

-->special permisssions allows user to acess command/file with owner permission

suid ->

Ex. We all know local user doesn't have permission to run "dmidecode" command. If the root user gave suid permission for this command then local user will able to run this command
It means no which user run this command command it will execute with root permissions
#Chmod u+s /sbin/dmidecode
s ----to allow acess
S ---to deny acess

sgid ->

if we give SGID permission to particular directory and if file created in that directory by root user or local user that files will get directory group ownership automatically.

```
[root@centos dir1]# chgrp dev demo
[root@centos dir1]# ll -d demo/
drwxr-xr-x. 2 root dev 6 Aug 12 10:10 demo/
[root@centos dir1]# chmod g+s demo
```

```
[root@centos dir1]# cd demo/
[root@centos demo]# touch file.txt
[root@centos demo]# ll
total 0
-rw-r--r--. 1 root dev 0 Aug 12 10:41 file.txt
```

sticky bit ->prevent file from deletion

If Sticky bit is enabled on a folder, the folder contents are deleted by only owner who created them and the root user, no one else can delete other users data in this folder

```
[root@centos dir1]# chmod 777 demo
```

```
[root@centos dir1]# chmod o+t demo
[root@centos dir1]# ll -d demo
drwxrwsrwt. 2 root dev 22 Aug 12 10:41 demo
[sam@centos dir1]$ rm -rvf file1.txt
rm: cannot remove 'file1.txt': Permission denied
```

ACL -Access Control List

ACL use to set permission over file and directory to specific user or specific group.

#getfacl --to check permission
#setfacl --to apply acl permission
#setfacl -m u:sam:rw file.txt

```
[root@centos demo]# setfacl -m u:sam:rw file.txt
```

-x to remove

```
[root@centos demo]# setfacl -x sam file.txt
```

Set acl to group dev:

```
[root@centos demo]# setfacl -m g:dev:rw file.txt
```

To remove group acl :

```
[root@centos demo]# setfacl -x g:dev file.txt
```

❖ Archiving and Compression

1. Archiving -->it is the process of combining multiple files and directories
tar ---> tape archive

#**tar** <options> <compress-filename>.tar <files/dir to be compressed>

options
-c --> create an archive
-f -->file name
-v -->view/verbose
-t -->list the content from archieve
-x -->extract
-C -->copy content from an archieve to another directory

#du -sh /etc ---check size of file/dir

1. archiving using tar
#tar -cvf /backup.tar /etc
#du -sh /backup.tar

2. listing archiving file
#tar -tvf /backup.tar

3. extract archive files
#tar -xvf /backup.tar ---extract at current location
#tar -xvf /backup.tar -C /demo/ ---extract at other dir
#du -sh /demo/etc

---

4. Compression -->it is the process of reducing the size of a file or directory
1. gzip >----fassssssssssst [gunzip]
2. bzip2 >----slow [bunzip2]
3. xz >---very slow [unzip]

#gzip -z <filename>.tar.gz
#bzip2 -j <filename>.tar.bzip2
#xz -J <filename>.tar.xz

Ex. #tar -cvf /backup.tar /etc
#du -sh /backup.tar
#gzip /backup.tar ---->compress backup.tar archive file using gzip
#du -sh /backup.tar.gz
#gunzip /backup.tar.gz ---->extract using gzip
---

3. Compression using tar command

#du -sh /etc
#tar -czvf /backup1.tar.gz /etc --->compress and archieve with gzip

extraction using tar command
#mkdir /backgz
#tar -xzvf /backup1.tar.gz -C /backgz ---->extract to other folder
---

❖ Package management

packages=software -> are the collection of programs.
dependency ->package requires shared library or another package

Que. How to install any package in linux?

->Low-level tool :[rpm]
manages package files installation, update and uninstallation
Red Hat ->rpm
Debian ->dpkg

->high-level tool:[yum]
install the package with their dependencies.
red hat->yum
Debian ->apt-get
mac-os ->brew

steps: open firefox
search for tree binary package
copy binary package url
on terminal type wget and paste the url using cntrl+shift+v
now install the package using rpm
#wget --> used to download package
#curl -->also download package [curl -o] and shows output

#rpm: It only install the package and not the dependency.

#rpm <option> <package-name>
-i install package file
-v verbose/view
-h show hash bar

-U Upgrade package
-q query package
-e erase package

#rpm –ivh tree-1.6.0-10.e17.x86_64.rpm

#yum: It download package with along with its dependencies

#yum <option/action> <package-name> [<-y/-d/-n>]
>install
>update
>list
>info
>remove
>history
>repolist
>clean ----clean cache

CREATE OWN REPOSITORY:
repository --> is a storage location which is used for installing and updating software packages

#yumdownloader httpd tree
#mkdir /my_packages
#createrepo -v /my_packages -->create metadata using createrepo command
#cd /etc/yum.repos.d
#vim myrepo.repo -->create repository with .repo extention
[server]
name=myrepo
baseurl=file:///my_packages --->path of metadata dir
enabled=1
gpgcheck=0

#yum clean all
#yum repolist all
#yum list tree
#yum install tree -y

#yum grouplist
#yum groupinstall groupname
#yum groupremove grpname

#yum autoremove

error:cant find base url for repo/7/x86_64
-->/etc/sysconfig/network-scriprt/ifcfg-enp03
onboot=yes

Process Management:

->process is nothing but the task that your machine is currently working on.

jobs->mostly started by user
daemons->starts with the system,and has root privileges which run in background

#jobs ->lists current jobs on terminal[options -r,-s,-p,-l]
ex. sleep 30 ->it will pause terminal for 30 seconds
to stop process[cntrl+z] to cancel[cntrl+c]
#kill %n --to terminate job
#bg > Display stopped or background jobs
ex. sleep 30 & ---& will execute command in bg

#fg > Brings the most recent background job to foreground
ex. fg %n ---bg to on screen

monitor process:
#ps >Display your currently running processes of system
#ps -ef

#top > Display and manage the processes
#uptime > Display load average
#w > Display load average

Niceness Value:

The niceness value is ranges from -20 to +19. Lower values indicate higher priority (higher CPU time), and higher values indicate lower priority (lower CPU time). The default niceness value is usually 0.

#nice -n 10 sleep
#ps -l
#renice -n 5 - p 4409 ---to change priority of already prioritized task

Process states:
running
stoped
zombie -->it is a process that has completed its execution[dead] but still has an entry in the process table[parent process]

kill :
ex. kill -9 4407 [pid] -> -9 signal number

Filter and search utility

cat
head
tail
more
less
grep ---grep “au” /country.txt , tail /etc/passwd |grep root
find
locate
which
pwd
sed ---stream editor
sed 's/old_word/new_word/g' input_file > output_file
sed -n '2p' country.txt

❖ Disk Partition

Disk partition is the method of dividing hard drive into multiple logical storage

ATA
defines how data is transferred between a computer's motherboard and storage devices, such as
*hard disk drives
*solid-state drives

[SATA] Serial Advanced Technology Attachment
->data is transferred one by one
[PATA] Parallel Advanced Technology Attachment
->multiple data bits at the same time
Virtual

Types of partition:

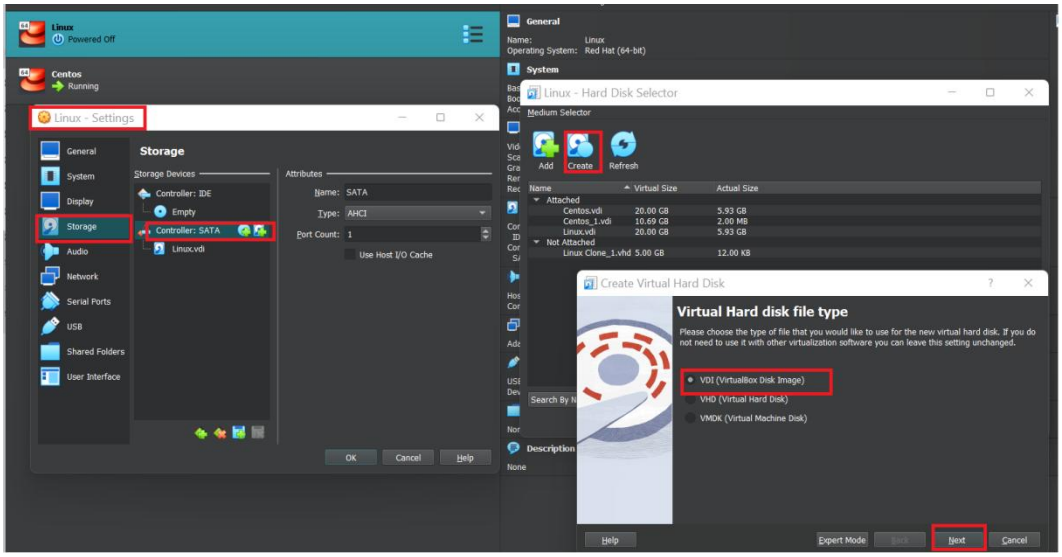
MBR -->master boot record [max partition->4=3 primary,1 extended]

GPT -->guid partition table [max partition->128]

Important commands:

lsblk -->listing blocks,shows system partition
fdisk -l -->disk detiail
df -hT -->check file type with mount point
partprobe --> save changes to kernel without reboot

Add new disk



Click on setting ->Storage->controller SATA
Click on + sign [new page will pop up]
Click on create -> select virtual disk image ->next
Choose disk -> attach to virtual Machine

Start the machine

#lsblk ->this command will list all the drives in your system
->you can see sdb disk is added to system

```
[root@localhost /]# lsblk
NAME            MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda               8:0    0   20G  0 disk
├sda1             8:1    0    1G  0 part /boot
└sda2             8:2    0   19G  0 part
  ├centos-root 253:0    0   17G  0 lvm  /
  └centos-swap 253:1    0    2G  0 lvm  [SWAP]
sdb               8:16   0 10.7G  0 disk
sr0              11:0    1 1024M  0 rom
```

#fdisk /dev/sdb -> this command will initiate new partition

```
[root@localhost /]# fdisk /dev/sdb
Welcome to fdisk (util-linux 2.23.2).

Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table
Building a new DOS disklabel with disk identifier 0xa922e183.

Command (m for help): m
```

Press m for help (it will show all the options related to disk partition)
n to create new partition
p print partition table
w save the changes
d delete partition

```
Command (m for help): n
Partition type:
   p   primary (0 primary, 0 extended, 4 free)
   e   extended
Select (default p): p
Partition number (1-4, default 1):
First sector (2048-22422991, default 2048):
Using default value 2048
Last sector, +sectors or +size{K,M,G} (2048-22422991, default 22422991): +5G
Partition 1 of type Linux and of size 5 GiB is set

Command (m for help): p

Disk /dev/sdb: 11.5 GB, 11480571904 bytes, 22422992 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk label type: dos
Disk identifier: 0xa922e183

   Device Boot      Start         End      Blocks   Id  System
/dev/sdb1            2048    10487807     5242880   83  Linux

Command (m for help): w
The partition table has been altered!
```

Select partition type
Press p for primary
Partition number ->just press enter
First sector ->just press enter
Last sector +size{K,M,G} -> +5G

Press m for help
Press p to print partition table
Press w to save

- Use **partprobe** command to save changes in kernel [it does not required reboot]
- Now **attached a file system** to a partion using mkfs command

```
[root@localhost /]# partprobe /dev/sdb
[root@localhost /]# mkfs.xfs /dev/sdb1
```

Mount the partition to dir [you can create new dir or use existing one]

#df -hT -->list partition along with file-system and mount point

```
[root@localhost /]# mount /dev/sdb1
mount: can't find /dev/sdb1 in /etc/fstab
[root@localhost /]# mkdir demo
[root@localhost /]# mount /dev/sdb1  /demo
[root@localhost /]# df -hT
Filesystem              Type      Size  Used Avail Use% Mounted on
devtmpfs                devtmpfs  903M     0  903M   0% /dev
tmpfs                   tmpfs     919M     0  919M   0% /dev/shm
tmpfs                   tmpfs     919M  9.5M  910M   2% /run
tmpfs                   tmpfs     919M     0  919M   0% /sys/fs/cgroup
/dev/mapper/centos-root xfs        17G  4.5G   13G  27% /
/dev/sda1               xfs      1014M  239M  776M  24% /boot
tmpfs                   tmpfs     184M   24K  184M   1% /run/user/1000
/dev/sdb1               xfs       5.0G   33M  5.0G   1% /demo
```

Mount drive permanently

```
[root@localhost /]# vim /etc/fstab
```

```
/dev/sdb1                                /demo   xfs     defaults  0 0
~
```

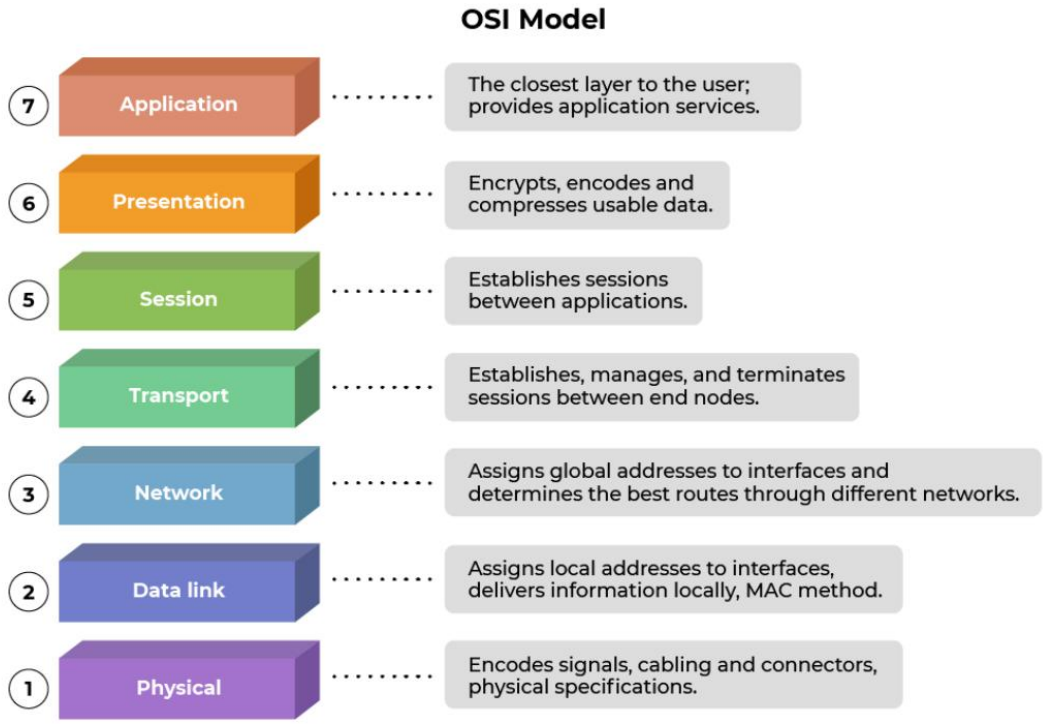
```
[root@localhost /]# mount -a
[root@localhost /]# df -hT
Filesystem              Type      Size  Used Avail Use% Mounted on
devtmpfs                devtmpfs  903M     0  903M   0% /dev
tmpfs                   tmpfs     919M     0  919M   0% /dev/shm
tmpfs                   tmpfs     919M  9.5M  910M   2% /run
tmpfs                   tmpfs     919M     0  919M   0% /sys/fs/cgroup
/dev/mapper/centos-root xfs        17G  4.5G   13G  27% /
/dev/sda1               xfs      1014M  239M  776M  24% /boot
tmpfs                   tmpfs     184M   28K  184M   1% /run/user/1000
/dev/sdb1               xfs       5.0G   33M  5.0G   1% /demo
```

❖ Networking

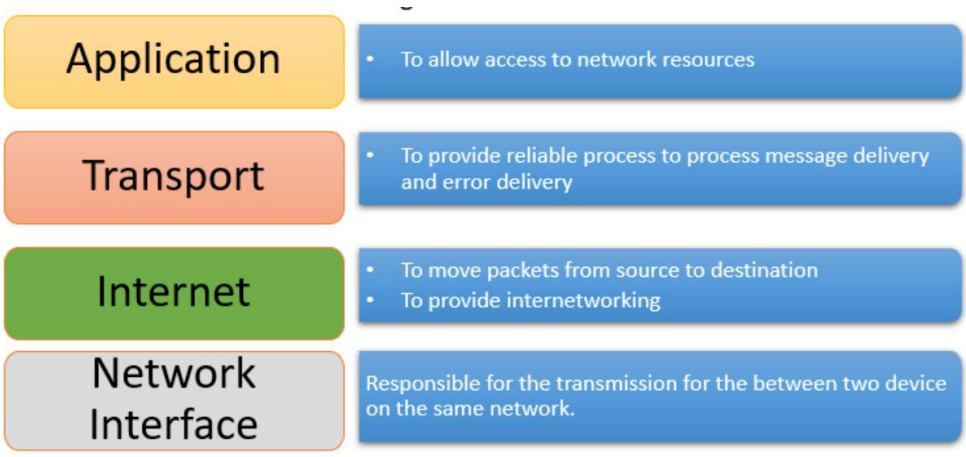
[1] OSI -open system interconnection

Que. Explain 7 layers of osi model ?

The OSI Model is a framework for how networking systems communicate and transmit data between sender and receiver.



[2]TCP/IP



#Networking Devices:

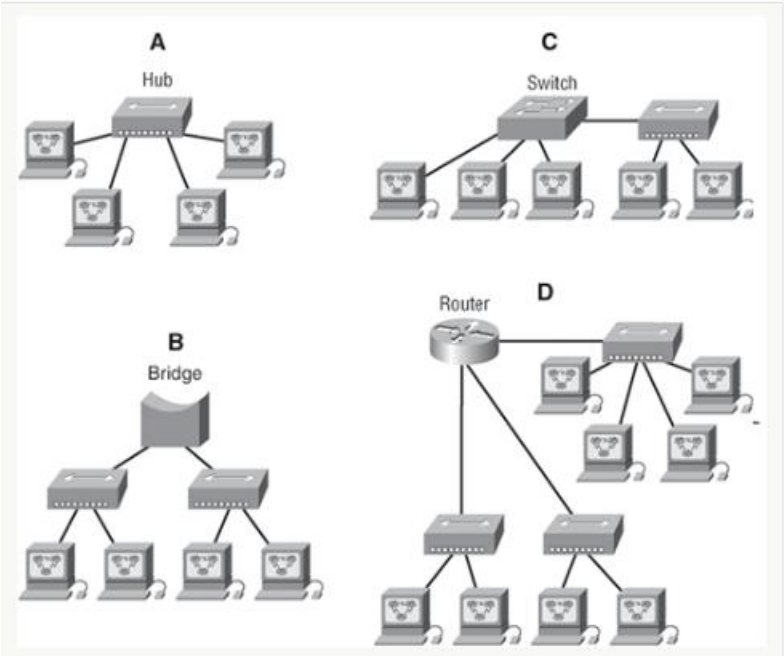
Hub, bridge, switch, and router are all networking devices used to connect and manage network traffic

1. Hub
   - A hub is a basic networking device that operates at the physical layer (Layer 1) of the OSI model.
   - It is essentially a multi-port repeater, which means it takes incoming data on one port and broadcasts it to all other ports.

2. Bridge
   - A bridge is a device that operates at the data link layer (Layer 2) of the OSI model.
   - It is used to filter and forward network traffic based on MAC (Media Access Control) addresses.

3. Switch
   - operates at the data link layer (Layer 2).
   - It is similar to a bridge but is typically more intelligent and has a larger number of ports.
   - Switches use MAC address tables to make forwarding decisions, allowing them to selectively forward traffic only to the appropriate port rather than broadcasting it to all ports.

4. Router
   - A router operates at the network layer (Layer 3) of the OSI model.
   - It is a more complex networking device that is used to connect different networks (e.g., local area networks or LANs) and make decisions about where to forward data based on IP (Internet Protocol) addresses.



In networking, a topology refers to the physical or logical layout of devices and the connections between them within a network.

1. Bus Topology
   - In a bus topology, all devices are connected to a single central cable (the bus).

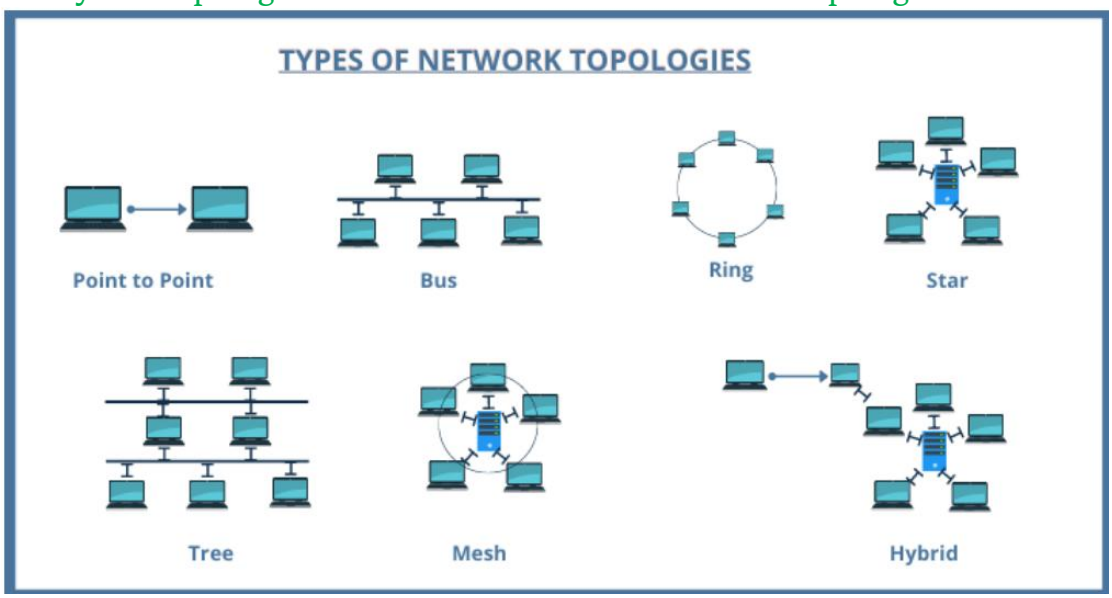
2. Star Topology
   - In a star topology, each device is connected directly to a central hub or switch
   - Star topologies are common in Ethernet LANs and are easy to manage.

3. Ring Topology
   - In a ring topology, devices are connected in a circular fashion, with each device connecting to exactly two other devices.
   - Data travels around the ring in one direction until it reaches its destination.

4. Mesh Topology
   - In a mesh topology, every device is connected to every other device in the network.
   - Mesh topologies provide redundancy and fault tolerance because multiple paths exist for data to travel.

5. Point-to-Point Topology
   - Point-to-point topology involves connecting each device to a single, dedicated link or channel.
   - This type of topology is commonly used for direct communication between two devices, such as in a WAN (Wide Area Network) link.

6. Hybrid Topology
- Hybrid topologies combine two or more different topologies.

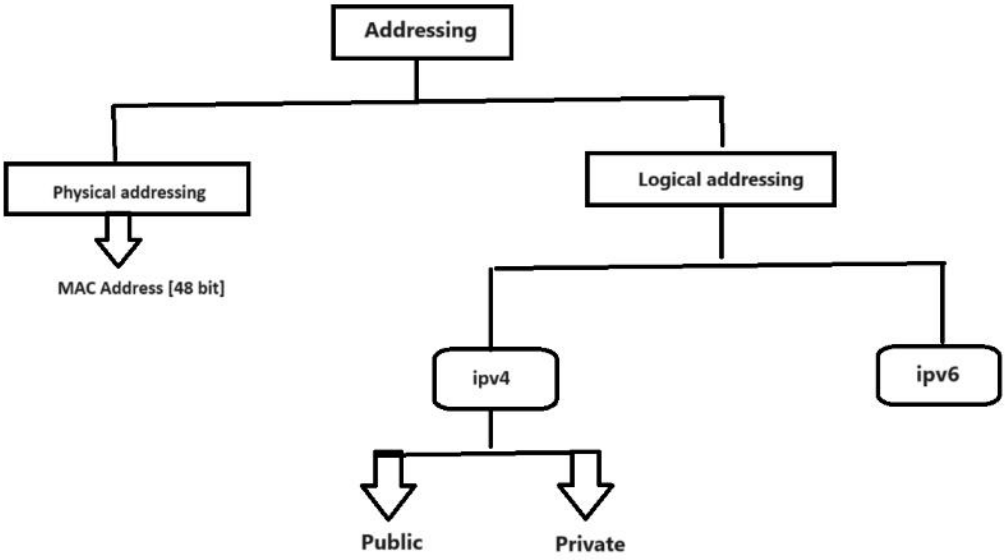


Types of computer networks:

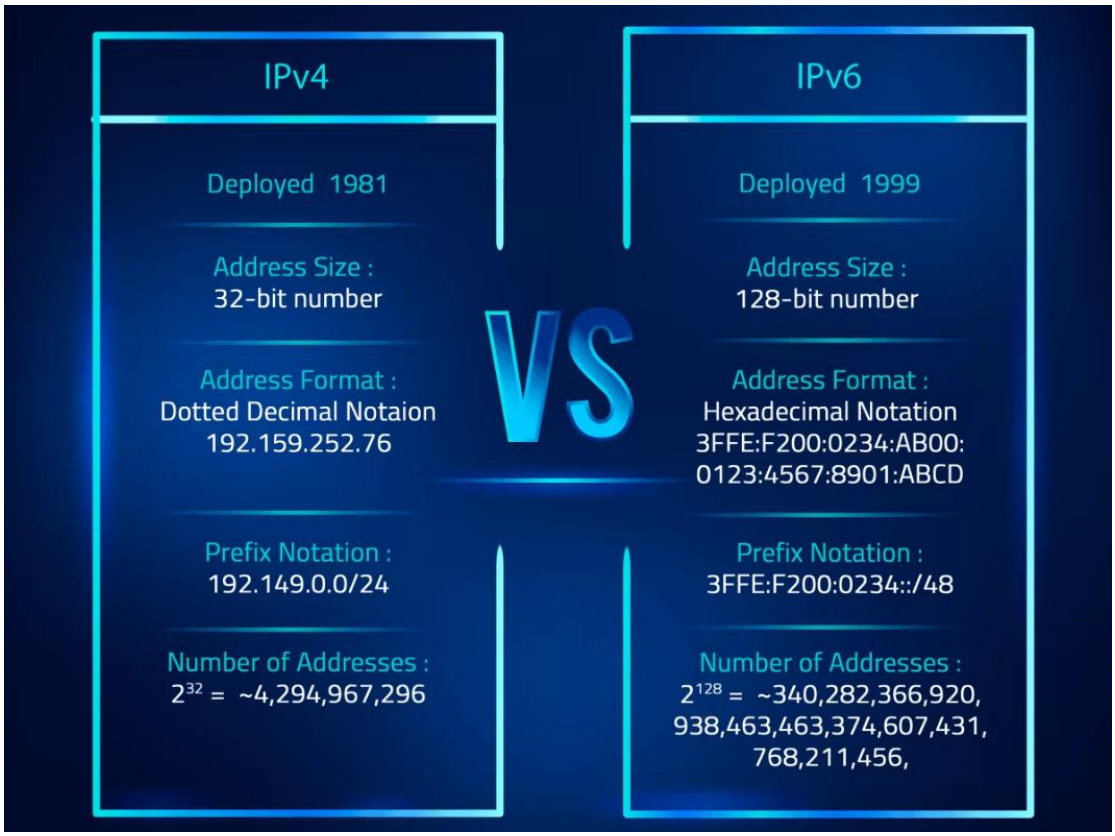
1. LAN (Local Area Network)
   - A LAN is a network that covers a small geographical area, such as a single building, office, or campus.
   - Common LAN technologies include Ethernet and Wi-Fi (Wireless LAN), and they are used to connect computers, servers, printers, and other devices within
     a localized environment.
2. MAN (Metropolitan Area Network)
   - A MAN is a network that covers a larger geographical area than a LAN but is still confined to a specific city.
   - MANs are used to connect multiple LANs within a city
3. WAN (Wide Area Network)
   - A WAN covers a much larger geographical area like cities, countries
   - The internet is a prime example of a global WAN, but private WANs are also used by organizations to connect their branch offices, data centers, and
     remote locations.
     eg. VPNs (Virtual Private Networks)

LAN	MAN	WAN
A computer network that interconnects computers within a limited area such as residence, office building, school or a laboratory	A computer network that interconnects user with computer resources in a geographical area larger than LAN but smaller than WAN	A computer network that extends over a large geographical area
Stands for Local Area Network	Stands for Metropolitan Area Network	Stands for Wide Area Network
Covers an area within 1km to 10km	Covers an area within 100km	Covers a large area that goes beyond 100km
Easier to design and maintain	Difficult and complicated to design and maintain	Difficult and complicated to design and maintain
High data transferring speed	Moderate data transferring speed	Low data transferring speed
Has a limited number of users - less congestion	Moderate congestion	Low congestion
A network in a home, school or an office	A network in a city or a small town	A network network covering a state or a country

ADDRESSING



Que. Difference between ipv4 and ipv6



Que. Difference between public and private ip

Public IP Address
1 Public IP Address is used to communicate outside the network.
2.Except private IP Addresses, rest IP addresses are public.
3.dynamic

Private IP Address
1. Private IP Address is used to communicate within the network.

2. Private IP Address range:
10.0.0.0 – 10.255.255.255,

172.16.0.0 – 172.31.255.255,

192.168.0.0 – 192.168.255.255

2. static

Classful
Que. Explain Ipv4-Classes:
Class A 1.0.0.0.0 - 126.0.0.0
Class B 128.0..0.0 - 191.255.0.0
Class C 192.0.0.0 - 223.255.255.0
Class D 224-239 -----research
Class E 240-255 ------research

Note: IP range 127.0.0.0 is reserved for loopback address[trouble shooting] which means an address that sends outgoing signals back to the same computer for testing

Network id ->It is a STARTING addressing of an IP [the first address is the network identification]
Broadcast id ->It is a LAST address of an IP [to transmit data to all of the hosts on the local subnet.]

Note:Both n/w id and bro. id are reserved by network [-2]

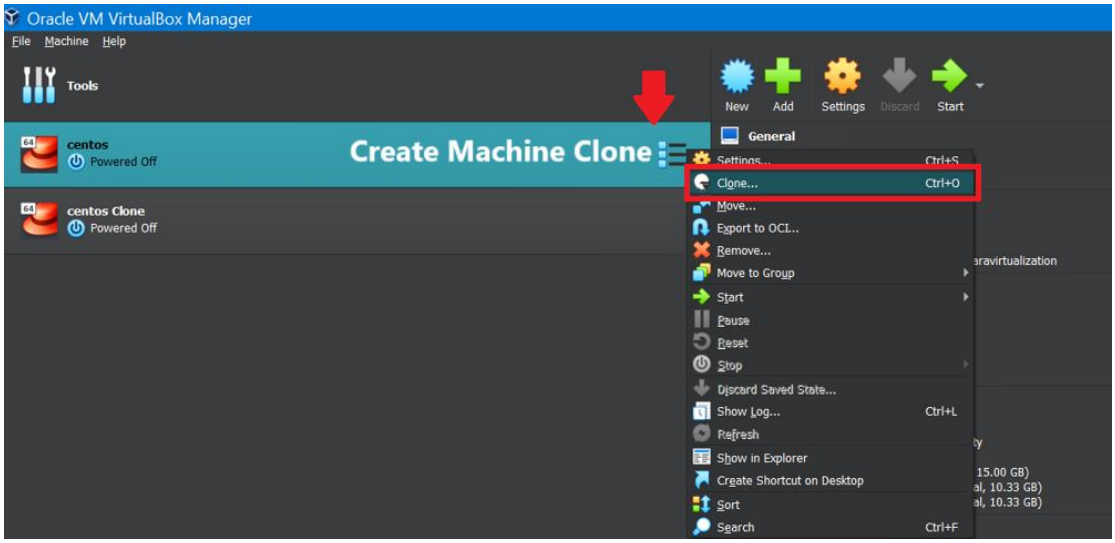
Subnet mask ->A subnet mask is for only internal usage within a network. Routers use subnet masks to route data packets to the right place

Ex. 192.168.1.0
Network id-192.168.1.0 ----host bit zero ----identify /represents network
Broadcast id-192.168.1.255 ----host bit 255 ----resvered to send msg to all
Subnet mask -255.255.255.0 ----network bits 255

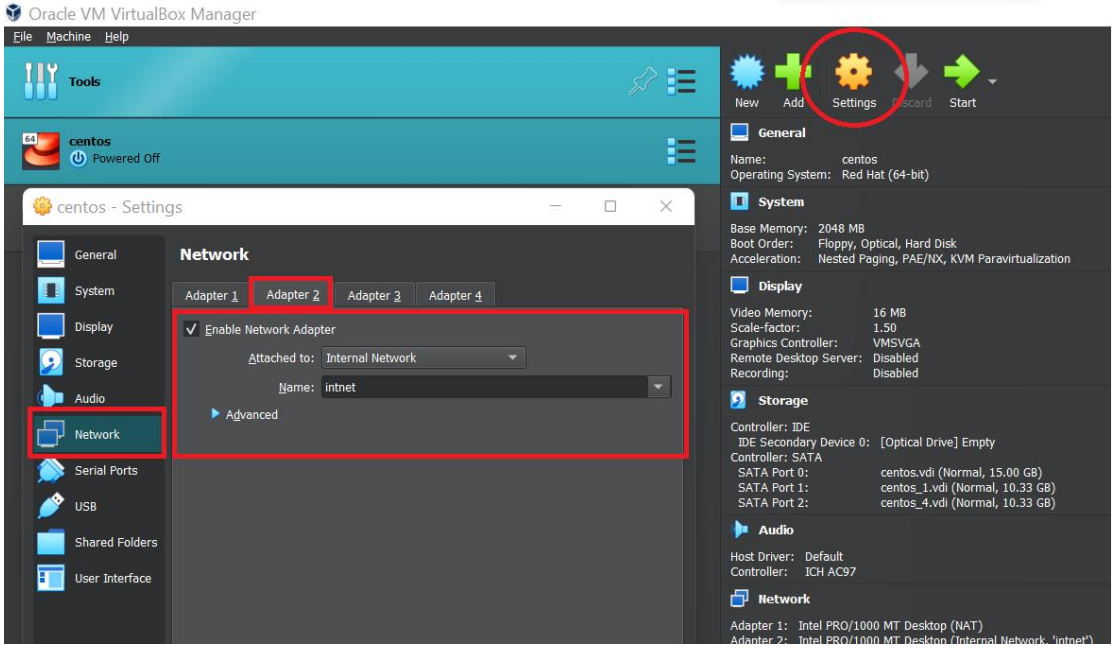
_Classless

#Sub-netting :helps you to breaks a big network into smaller parts, making it easier to manage, control, and use the available resources efficiently.
steps:
1)convert ip into binary
2) i) no of n/w = 2^n number of on bits
ii) no of usable host=2^total number of off bits [total zeros]
3)find subnet with magical number
ex. 1 1 0 0 0 0 0 here add two on bits and substract from max bits
256-192=64
4) subnet
5) Subnet range

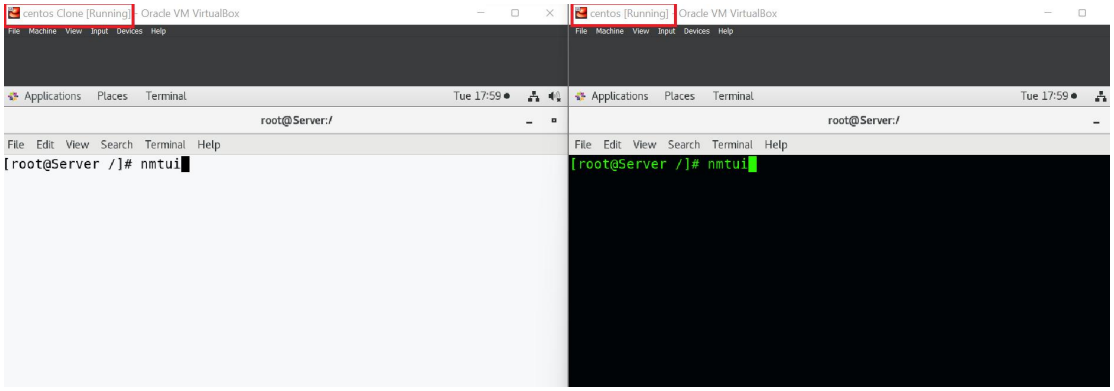
Adding a network adapter to a virtual machine

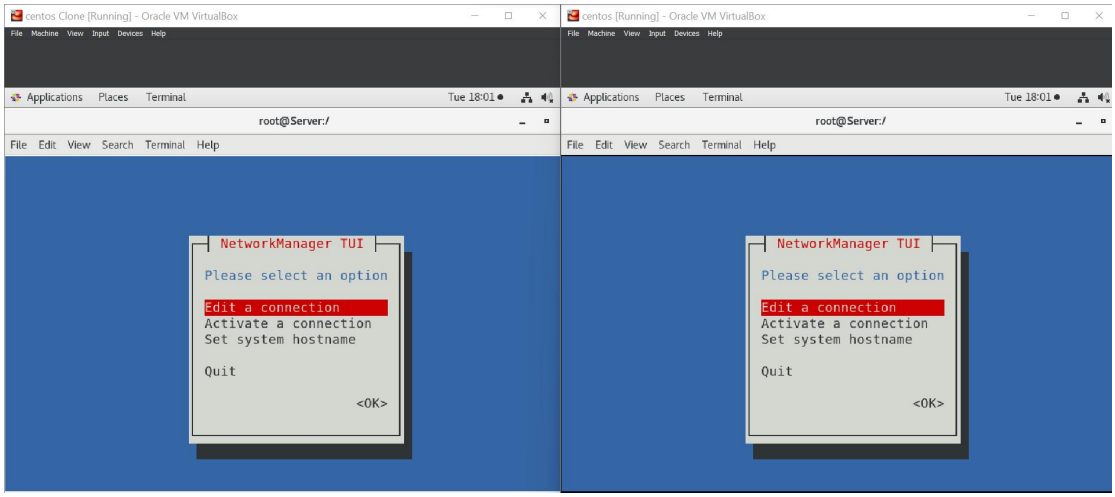


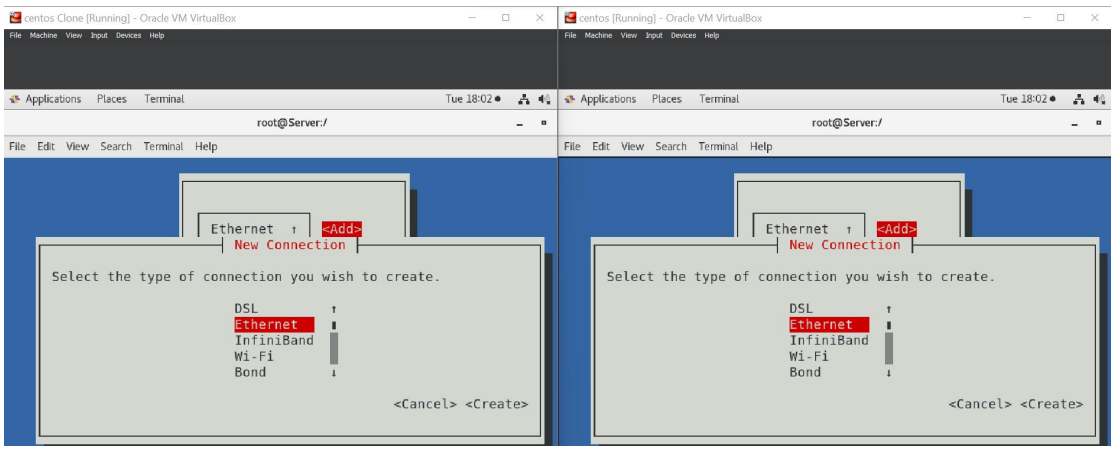
Choose Internal N/W:
Internal Network: It creates a private network between the host and the VMs.



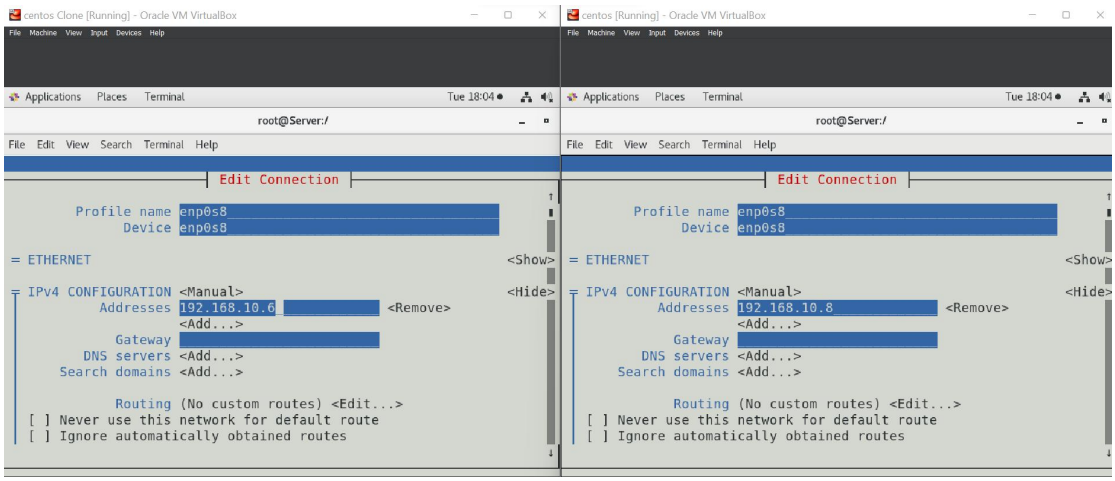
Start the machines
Go to Network Manager Using nmtui command

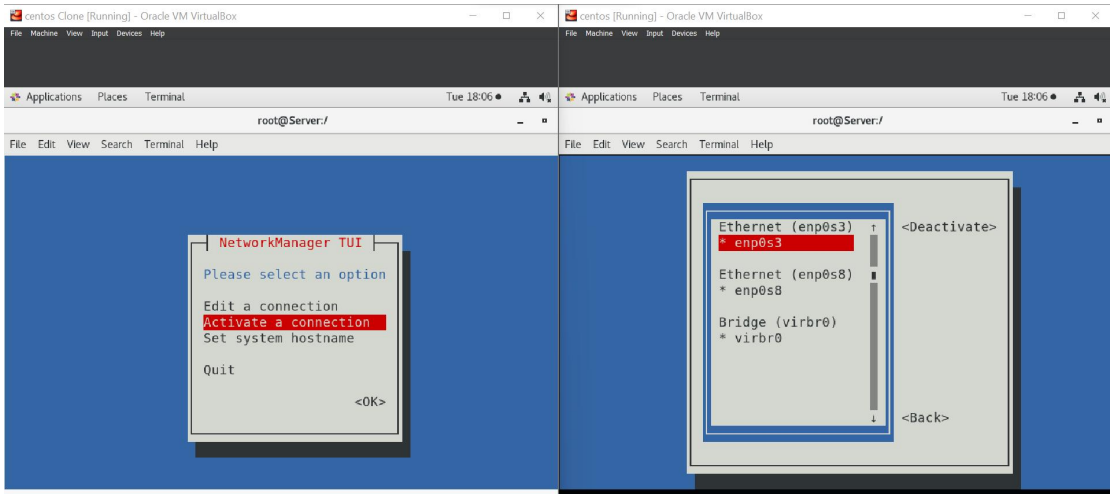




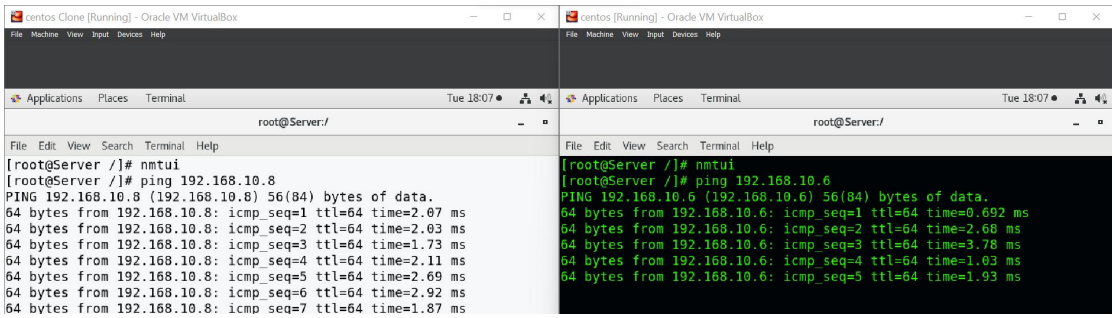


Make Sure ipv4 addresses must be of same range





Ping Clone ip into main server and vice versa to test network connectivity



❖ Web Server

A web server stores and delivers web content to users over the internet. It handles requests from web browsers, retrieves web pages, and sends them to the users' devices. The most common protocol for communication is HTTP. Examples of web servers include Apache, Nginx, and Microsoft IIS.

Apache / Httpd

#yum install httpd -y
#systemctl start httpd
#systemctl enable httpd
#systemctl status httpd ---> active

#echo " welcome to cloudblitz" > */var/www/html/index.htm*l

open browser:

type- localhost
it will show content in the index.html file
"welcome to cloudblitz"

NGINX server:
yum update
yum install epel-release -y ----download extra packages
yum install nginx
systemctl start nginx
systemctl enable nginx
systemctl status nginx

echo " welcome to nginx" >*/usr/share/nginx/html* ----default dir for web pages
/etc/nginx/conf.d/default.conf. -------default server configuration
yum
note: if error delete httpd
add ports:
firewall-cmd --zone=public --add-service=http --permanent
firewall-cmd --zone=public --add-https

Appache Tomcat server

Port->8080
download java [1.8]
yum install java-1.8* -y

installation of tomcat

download tomcat file
tomcat official site --> tomcat 8 --> copy zip file link --> on terminal download using
wget https://dlcdn.apache.org/tomcat/tomcat-8/v8.5.93/bin/apache-tomcat-8.5.93.zip

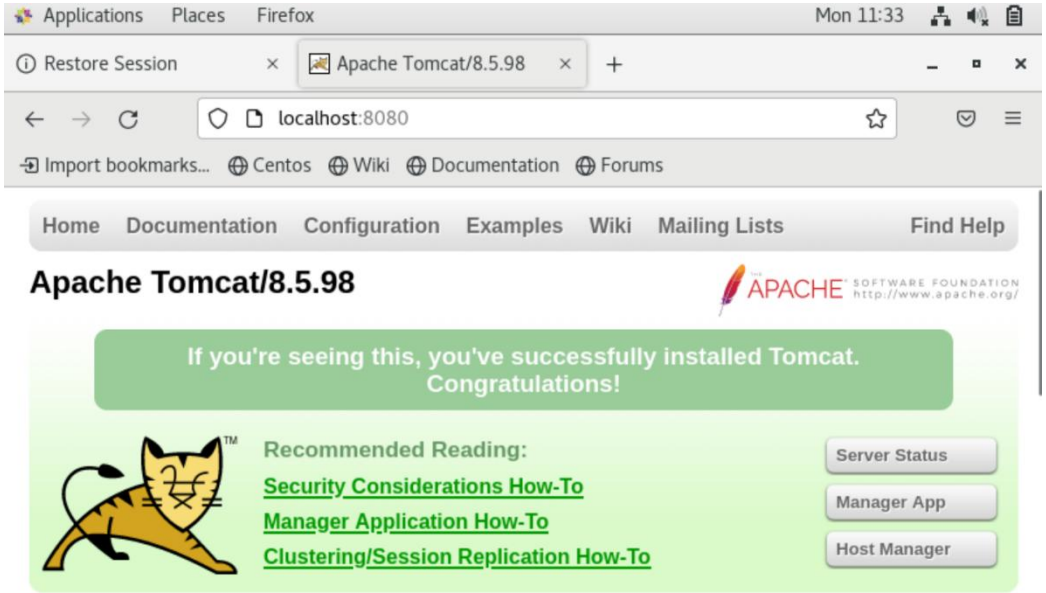
unnzip the file
unzip apache-tomcat-8.5.93.zip
cd apache-tomcat-8.5.93

Assign Permission
cd ***bin***

make sh file executable (permission)
chmod +x *.sh or chmod +x catalina.sh

tomcat start stop command
./catalina.sh stop
./catalina.sh start

Go to browser
localhost:8080 or ip:8080



❖ Services

Mariadb:

what is data?
->data means information. it could be names, numbers, dates, or any other
kind of information
what is database?
->a database is like a place where we store and organization's information

Types:
1. SQL -structured query language/ Relational
->SQL requires predefined schema[structured]
->SQL databases are well-suited for applications with structured data and complex relationships, such as financial systems, e-commerce platforms, and systems that requires data consistency.
->Examples: MySQL, PostgreSQL, Oracle Database, Microsoft SQL Server
2.NoSQL- non-relational
->NoSQL requires dynamic schema[unstructured]

->NoSQL databases are suitable for applications with rapidly changing or unstructured data, such as social media platforms, content management systems, real-time analytics
->Examples: MongoDB, DynamoDB, Cassandra, Redis.

Installation steps:
yum install mariadb-server -y
vim /etc/my.cnf ---configuration file[skip-networking=1 --cant acess remotely]
systemctl start mariadb
systemctl enable mariadb
systemctl status mariadb
firewall-cmd --add-service=mysql --permanent
firewall-cmd --reload

mysql_secure_installation
..set root password for mariadb
->just press enter
->set root passwd ->y
->-y
->-y
mysql -u root -p -----login into server
enter password:
Basic database commands:
>show databases;
>create database database_name;
>use database_name;
>create table student_info (Name varchar(20), RollNo int, City Varchar(20));
>show tables
>describe student_info
>insert info student_info values ('user1',22,'Nagpur');
>select * from student_info; ---view data from table
>select name from student_info where city='nagpur'

❖ BOOTING PROCEDURE

1. BIOS – Basic input/output System
first program to execute[POST-power on self test- all working properly] check bootable device os load
kernel

2. MBR – Master Boot Record [512 bytes]
boot loader [which partition]
partition table

3. GRUB2 – Grand Unified Boot loader
path --> / boot/grub2/grub.cfg
load kernel into memory[RAM]

4. Kernel – Kernel
mount partitions
5. Systemd – First process of system start all the necessary processes
/etc/systemD/system/default.target
init/systemD
#ps -ef|grep systemd

❖ Firewall

A firewall is a network security device or software that monitors and controls incoming and outgoing network traffic based on predetermined security rules.

#systemctl stop firewalld
#systemctl start firewalld
#firewall-cmd --add-service=httpd --permanent

firewall-cmd --help | grep forward
->firewall-cmd --list-all
->firewall-cmd --add-forward-port=port=80:proto=tcp:toport=8888 --permanent
->systemctl restart firewalld

->->firewall-cmd --list-all

❖ Log Management:

```
[root@ip-172-31-24-16 ~]# tail -5 /var/log/messages
Jan 30 04:43:47 ip-172-31-24-16 amazon-ssm-agent: </body>
Jan 30 04:43:47 ip-172-31-24-16 amazon-ssm-agent: </html>
Jan 30 04:45:08 ip-172-31-24-16 dhclient[3012]: XMT: Solicit
on eth0, interval
131720ms.
```

Date and time: Every log message starts with a timestamp. For filtering purposes,
Host: The host the message originated from. This is relevant because rsyslogd can be
configured to handle remote logging as well.
Service or process name: The name of the service or process that generated the message.
Message content: The content of the message, which contains the exact message that has
been logged.

Questions Linux:

1. What is the Linux operating system, and how does it differ from other operating systems?
2. Describe the role of the Linux kernel in the operating system.
3. Name a few popular Linux distributions and their specific use cases.
4. Describe the directory structure of Linux.
5. What is the root directory in Linux, and why is it significant?
6. How do you list files and directories in a directory using the command line?
7. Explain the purpose of the ls, cd, pwd, and mkdir commands.
8. File Permissions and Security:
9. Explain the concept of file permissions in Linux.
10. How do you change file permissions using numeric notation and symbolic notation?
11. What is the umask command used for?
12. How can you change the ownership of a file or directory using the command line?
13. What is the purpose of the sudo command, and how is it used to manage permissions?

14. Process Management:

15. How do you list currently running processes on a Linux system?
16. Explain the significance of the process ID (PID) in Linux.
17. How do you terminate a running process using the command line?
18. What is the purpose of the ps and top commands?

Networking and Services:
19. How can you find the IP address of a Linux system?
20. What is the purpose of the ping command, and how does it work?
21. System Monitoring and Performance:
22. How can you check CPU and memory usage on a Linux system?
23. Explain the purpose of the top command.
24. What is the load average?
25. System Logs:
26. Where are system log files typically stored in Linux?
27. Explain the purpose of the dmesg command.
28. How can you view the contents of log files using the command line?
29. SSH and Security:
30. What is SSH, and how does it provide secure communication?
31. How do you generate SSH key pairs, and why are they useful?
32. Explain the process of securing SSH access using key-based authentication.
33. Describe the purpose of the /etc/fstab file.
34. How do you list available disks and partitions on a Linux system?
35. Explain the steps to mount and unmount a filesystem using the command line.
36. What is the purpose of the du and df commands?
37. What is virtualization?
38. What is a web server, and what are some popular web servers in Linux?

User Management:
39. How do you create a new user in Linux using the command line?
40. Explain the purpose of the `/etc/passwd` file and its fields.
41. What is the difference between the user's login name (username) and the user's full name (gecos field)?
42. How can you set a password for a user account using the command line?
43. Describe the purpose of the `/etc/shadow` file in Linux.
44. How do you lock and unlock a user account?
45. What is the `useradd` command, and what are its common options?
46. How can you modify the properties of an existing user account using the `usermod` command?
47. Explain the concept of the default shell for a user account and how you can change it.
48. What is the `passwd` command used for, and how can a user change their password?

49. Group Management
50. What is the purpose of user groups in Linux?
51. How do you create a new group using the `groupadd` command?
52. How can you add a user to an existing group using the `usermod` command?
53. Explain the difference between a primary group and a supplementary group.
54. How can you list the groups a user belongs to using the `groups` command?
55. Describe the significance of the `/etc/group` and `/etc/gshadow` files in group management.
56. What is the purpose of the `newgrp` command, and how is it used?
57. How can you change the primary group of a user using the `usermod` command?
58. Explain the concept of group ownership of files and directories in Linux.
59. How do you grant read, write, and execute permissions to a group for a specific file?

60. Basic Permissions:
61. What are the three basic permissions in Linux file systems?
62. Explain the meanings of read, write, and execute permissions for files and directories.
63. How can you view the permissions of a file using the ls command?
64. Describe the output of the ls -l command and how it displays permissions.

65. Default Permissions and umask:
66. What is the default permission mode for newly created files and directories?
67. Diff. Apache and nginx
68. Diff tcp and udp protocol
69. Question set:
70. Que.What is ACL? And how to set acl?
71. Que. What is special permission and its types?
72. Que. What is Umask ? what is default umask for root and local user?
73. Que. What is Default Permission for root and local user?
74. Que.How to change owner and group owner?
75. Que. Diff between Hard link and soft link
76. Que. What is the user id of a root user ?
77. Que. How do you Assign Admin to group?
78. Que. Exxplain the Fields of /etc/gshadow
79. Que.what are the Fields of /etc/group ?
80. Que. Explain password policy ?
81. Que. What are the Fields of /etc/passwd ?

82. Que. What happens when we add new user to system?
83. Que. What are the Re-directors used in linux?
84. Que. What is vi/vim and explain modes of vi?
85. Que. How many types of Users are there in linux ? explain ?
86. Que. What are various Distributions /Versions /Flavors of Linux ?
87. Que. Explain the FEATURES of Linux
88. Que. differences between windows and linux ?
89. Que . How to I check the available shells ?
90. Que . What is the command to change shell ?
91. Que . How to check current running shell ?
92. Que. Define shell and explain types of shells ?
93. Que. Explain the Architecture of Linux
94. Que.Can you explain what is Operating System

95. Que. how to extract archive files?
96. Que. what are the file compression methods in linux?
97. Que. Which is the low level tool to download any package in linux?
98. Que.Which is the High level tool to download any package in linux?
99. Que. What is diff between yum and rpm?
100. Que. diff between wget and curl command
101. Que.how many max partition can be done in mbr?
102. Que.What are the types of mounting?
103. Que. How to mount drive permanently?
104. Que.What is lsblk command does?
105. Que.what is the command to check mounting point and file system
106. Que. how to check size of perticular directory or file ?
107. Que. explain 7 layers of osi model
108. Que. Difference between OSI and TCP/IP
109. Que. Difference between public and private ip
110. Que. Difference between ipv4 and ipv6
111. Que. Explain Ipv4-Classes?
112. Que. What is the range of private ip addresses
113. Que. Wht is network id, broadcast id and subnet mask